
Chapter 6. Signal Processing Basics

This chapter provides insights about several lock-in amplifier principles not necessarily linked to a specific instrument from Zurich Instruments. Since the appearance of the first valve-based lock-in amplifiers in the 1930s the physics have not changed, but the implementation and the performance have evolved greatly. Many good lock-in amplifier primers have appeared in the past decades, and some of them appear outdated now because they were written with analog instruments in mind. This section does not aim to replace any existing primer, but to complete them with a preferred emphasis on digital lock-in amplifiers.

The first subsection describes the principles of lock-in amplification, followed by the description of the function of discrete-time filters. After, we discuss the definition of the full range sensitivity, a specification parameter particularly important for analog lock-in amplifiers but with somewhat reduced importance for digital instruments. In the following, we describe the function and use of sinc filtering in particular for low-frequency lock-in measurements. The last section is dedicated to the zoom FFT feature. Innovative in the context of lock-in amplifiers, zoom FFT offers a fast and high-resolution spectral analysis around the lock-in operation frequency.

6.1. Principles of Lock-in Detection

Lock-in demodulation is a technique to measure the amplitude A_s and the phase Θ of a periodic signal with the frequency $\omega_s = 2\pi f_s$ by comparing the signal to a reference signal. This technique is also called phase-sensitive detection. By averaging over time the signal-to-noise ratio (SNR) of a signal can be increased by orders of magnitude, allowing very small signals to be detected with a high accuracy making the lock-in amplifier a tool often used for signal recovery. For both signal recovery and phase-sensitive detection, the signal of interest is isolated with narrow band-pass filtering therefore reducing the impact of noise in the measured signal.

Figure 6.1 shows a basic measurement setup: a reference V_r signal is fed to the device under test. This reference signal is modified by the generally non-linear device with attenuation, amplification, phase shifting, and distortion, resulting in a signal $V_s = A_s \cos(\omega_s t + \Theta_s)$ plus harmonic components.

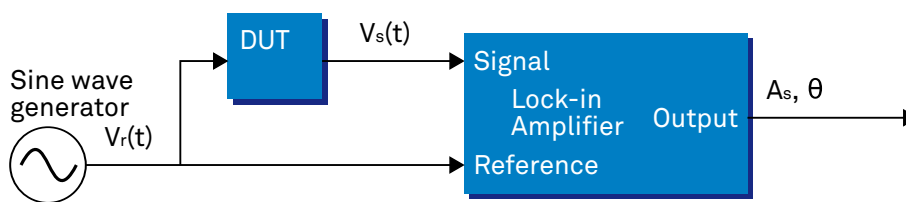


Figure 6.1. Basic measurement setup incorporating a lock-in amplifier

For practical reasons, most lock-in amplifiers implement the band-pass filter with a mixer and a low-pass filter (depicted in Figure 6.2): the mixer shifts the signal of interest into the baseband, ideally to DC, and the low-pass filter cuts all unwanted higher frequencies.

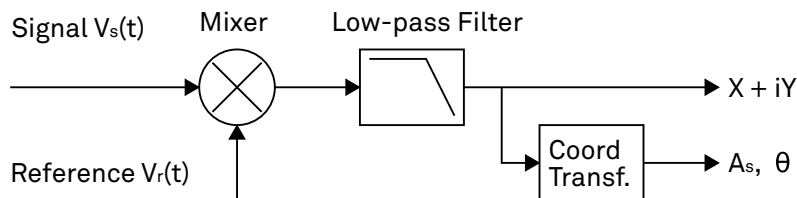


Figure 6.2. Mixing and low-pass filtering performed by the lock-in amplifier

The input signal $V_s(t)$ is multiplied by the reference signal $V_r(t) = \sqrt{2} e^{-i\omega_r t}$, where $\omega_r = 2\pi f_r$ is the demodulation frequency and i is the imaginary unit. This is the complex representation of a sine and cosine signal (phase shift 90°) forming the components of a quadrature demodulator, capable of measuring both the amplitude and the phase of the signal of interest. In principle it is possible to multiply the signal of interest with any frequency, resulting in a heterodyne operation. However the objective of the lock-in amplifier is to shift the signal as close as possible to DC, therefore the frequency of the reference and the signal is chosen similar. In literature this is called homodyne detection, synchrodyne detection, or zero-IF direct conversion.

The result of the multiplication is the signal

$$V_s(t) \cdot V_r(t) = V_s(t) \cdot \sqrt{2} e^{-i\omega_r t} = \frac{A_s}{\sqrt{2}} e^{+i[(\omega_s - \omega_r)t + \Theta]} + \frac{A_s}{\sqrt{2}} e^{-i[(\omega_s + \omega_r)t + \Theta]}$$

Equation 6.1. Multiplication of signal of interest with reference signal

It consists of a slow component with frequency $\omega_s - \omega_r$ and a fast component with frequency $\omega_s + \omega_r$.

The demodulated signal is then low-pass filtered with an infinite impulse response (IIR) RC filter, indicated by the symbol $\langle \cdot \rangle$. The frequency response of the filter $F(\omega)$ will let pass the low frequencies $F(\omega_s - \omega_r)$ while considerably attenuating the higher frequencies $F(\omega_s + \omega_r)$. Another way to consider the low-pass filter is an averager.

$$X + iY = \langle V_s(t) \cdot \sqrt{2} e^{-i\omega_r t} \rangle \approx F(\omega_s - \omega_r) \frac{A_s}{\sqrt{2}} e^{i[(\omega_s - \omega_r)t + \Theta]}$$

Equation 6.2. Averaging the result of the signal multiplication

The result after the low-pass filter is the demodulated signal $X + iY$, where X is the real and Y is the imaginary part of a signal depicted on the complex plane. These components are also called in-phase and quadrature components. The transformation of X and Y into the amplitude R and phase Θ information of $V_s(t)$ can be performed with trigonometric operations.

It is interesting to note that the value of the measured signal corresponds to the RMS value of the signal, which is equivalent to $R = A_s/\sqrt{2}$.

Most lock-in amplifiers output the values (X, Y) and (R, Θ) encoded in a range of -10 V to $+10$ V of the auxiliary output signals.

6.1.1. Lock-in Amplifier Applications

Lock-in amplifiers are employed in a large variety of applications. In some cases the objective is measuring a signal with good signal-to-noise ratio, and then that signal could be measured even with large filter settings. In this context the word phase sensitive detection is appropriate. In other applications, the signal is very weak and overwhelmed by noise, which forces to measure with very narrow filters. In this context the lock-in amplifier is employed for signal recovery. Also, in another context, a signal modulated on a very high frequency (GHz or THz) that cannot be measured with standard approaches, is mixed to a lower frequency that fits into the measurement band of the lock-in amplifier.

One example for measuring a small, stationary or slowly varying signal which is completely buried in the $1/f$ noise, the power line noise, and slow drifts. For this purpose a weak signal is modulated to a higher frequency, away from these sources of noise. Such signal can be efficiently mixed back and measured in the baseband using a lock-in amplifier. In [Figure 6.3](#) this process is depicted. Many optical applications perform the up-mixing with a chopper, an electro-optical modulator, or an acousto-optical modulator. The advantage of this procedure is that the desired signal is measured in a spectral region with comparatively little noise. This is more efficient than just low-pass filtering the DC signal.

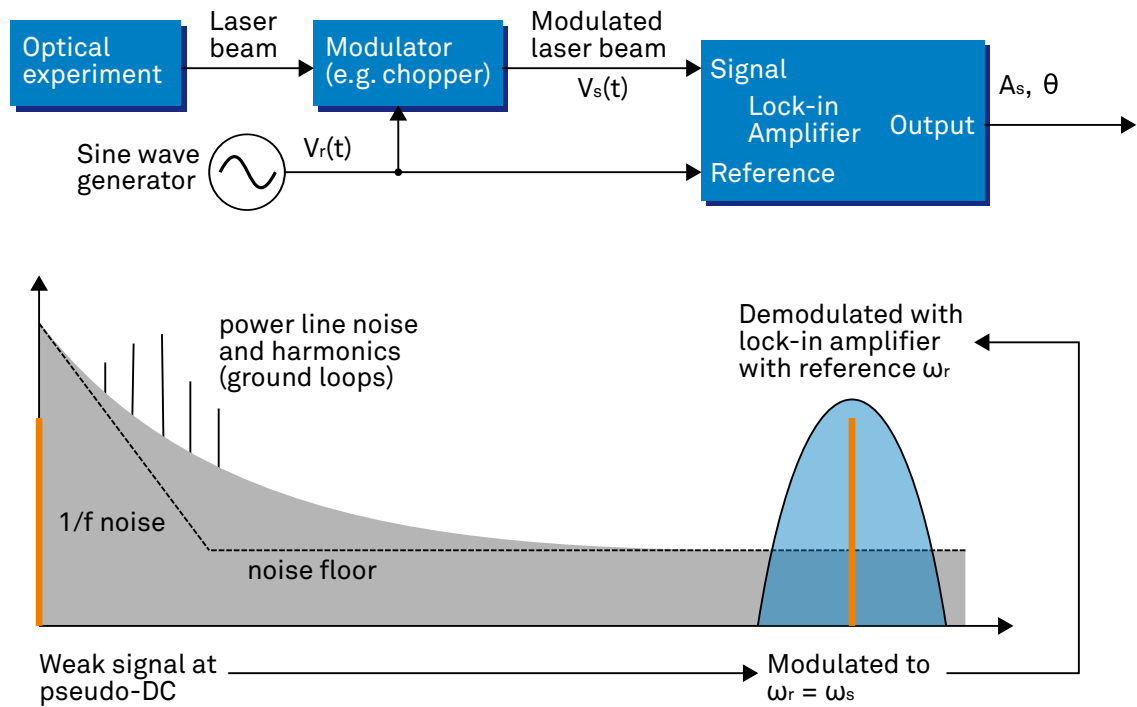


Figure 6.3. Lock-in measurement of a noisy DC signal

6.2. Signal Bandwidth

The signal bandwidth (BW) theoretically corresponds to the highest frequency components of interest in a signal. In practical signals, the bandwidth is usually quantified by the cut-off frequency. It is the frequency at which the transfer function of a system shows 3 dB attenuation relative to DC ($BW = f_{\text{cut-off}} = f_{-3\text{dB}}$); that is, the signal power at $f_{-3\text{dB}}$ is half the power at DC. The bandwidth, equivalent to cut-off frequency, is used in the context of dynamic behavior of a signals or separation of different signals. This is for instance the case for fast-changing amplitudes or phase values like in a PLL or in imaging applications, or when signals closely spaced in frequency need to be separated.

The noise equivalent power bandwidth (NEPBW) is also a useful figure, and it is distinct from the signal bandwidth. This unit is typically used for noise measurements: in this case one is interested in the total amount of power that passes through a low-pass filter, equivalent to the area under the solid curve in Figure 6.4. For practical reasons, one defines an ideal brick-wall filter that lets pass the same amount of power under the assumption that the noise has a flat (white) spectral density. This brick-wall filter has transmission 1 from DC to f_{NEPBW} . The orange and blue areas in Figure 6.4 then are exactly equal in a linear scale.

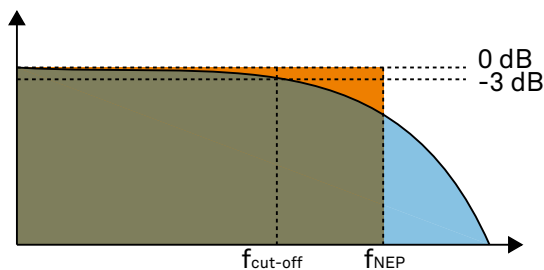


Figure 6.4. Signal bandwidth and noise equivalent power bandwidth

It is possible to establish a simple relation between the $f_{\text{cut-off}}$ and the f_{NEPBW} that only depends on the slope (or roll-off) of the filter. As the filter slope actually depends on the time constant (TC) defined for the filter, it is possible to establish the relation also to the time constant. It is intuitive to understand that for higher filter orders, the $f_{\text{cut-off}}$ is closer to the f_{NEPBW} than for smaller orders.

The time constant is a parameter used to interpret the filter response in the time domain, and relates to the time it takes to reach a defined percentage of the final value. The time constant of a low-pass filter relates to the bandwidth according to the formula

$$TC = \frac{FO}{2\pi f_{\text{cut-off}}} \quad (6.3)$$

where FO is said factor that depends on the filter slope. This factor, along with other useful conversion factors between different filter parameters, can be read from the following table.

Table 6.1. Summary of conversion factors for bandwidth definitions

filter order	filter roll-off	FO	$f_{\text{cut-off}}$	f_{NEPBW}	$f_{\text{NEPBW}} / f_{\text{cut-off}}$
1 st	6 dB/oct	1.0000	$0.1592 / TC$	$0.2500 / TC$	1.5708
2 nd	12 dB/oct	0.6436	$0.1024 / TC$	$0.1250 / TC$	1.2203
3 rd	18 dB/oct	0.5098	$0.0811 / TC$	$0.0937 / TC$	1.1554
4 th	24 dB/oct	0.4350	$0.0692 / TC$	$0.0781 / TC$	1.1285
5 th	30 dB/oct	0.3856	$0.0614 / TC$	$0.0684 / TC$	1.1138

filter order	filter roll-off	FO	$f_{\text{cut-off}}$	f_{NEPBW}	$f_{\text{NEPBW}} / f_{\text{cut-off}}$
6 th	36 dB/oct	0.3499	0.0557 / TC	0.0615 / TC	1.1046
7 th	42 dB/oct	0.3226	0.0513 / TC	0.0564 / TC	1.0983
8 th	48 dB/oct	0.3008	0.0479 / TC	0.0524 / TC	1.0937

6.3. Discrete-Time Filters

6.3.1. Discrete-Time RC Filter

There are many options how to implement digital low-pass filters. One common filter type is the exponential running average filter. Its characteristics are very close to those of an analog resistor-capacitor RC filter, which is why this filter is sometimes called a discrete-time RC filter. The exponential running average filter has the time constant $TC = \tau_N$ as its only adjustable parameter. It operates on an input signal $X_{in}[n, T_S]$ defined at discrete times $nT_S, (n+1)T_S, (n+2)T_S$, etc., spaced at the sampling time T_S . Its output $X_{out}[n, T_S]$ can be calculated using the following recursive formula,

$$X_{out}[n, T_S] = e^{-T_S/\tau_N} X_{out}[n-1, T_S] + (1 - e^{-T_S/\tau_N}) X_{in}[n, T_S]$$

Equation 6.4. Time domain response of the discrete-time RC filter

The response of that filter in the frequency domain is well approximated by the formula

$$H_1(\omega) = \frac{1}{1 + i \cdot \omega \cdot \tau_N}$$

Equation 6.5. Frequency domain response of the first-order discrete-time RC filter

The exponential filter is a first-order filter. Higher-order filters can easily be implemented by cascading several filters. For instance the 4th order filter is implemented by chaining 4 filters with the same time constant $TC = \tau_N$ one after the other so that the output of one filter stage is the input of the next one. The transfer function of such a cascaded filter is simply the product of the transfer functions of the individual filter stages. For an n-th order filter, we therefore have

$$H_n(\omega) = \frac{1}{(1 + i \cdot \omega \cdot \tau_N)^n}$$

Equation 6.6. Frequency domain response of the n-th order discrete-time RC filter

The attenuation and phase shift of the filters can be obtained from this formula. Namely, the filter attenuation is given by the absolute value squared $|H_n(\omega)|^2$. The filter transmission phase is given by the complex argument $\arg[H_n(\omega)]$.

6.3.2. Filter Settling Time

The low-pass filters after the demodulator cause a delay to measured signals depending on the filter order and time constant $TC = \tau_N$. After a change in the signal, it will therefore take some time before the lock-in output reaches the correct measurement value. This is depicted in [Figure 6.5](#) where the response of cascaded filters to a step input signal this is shown.

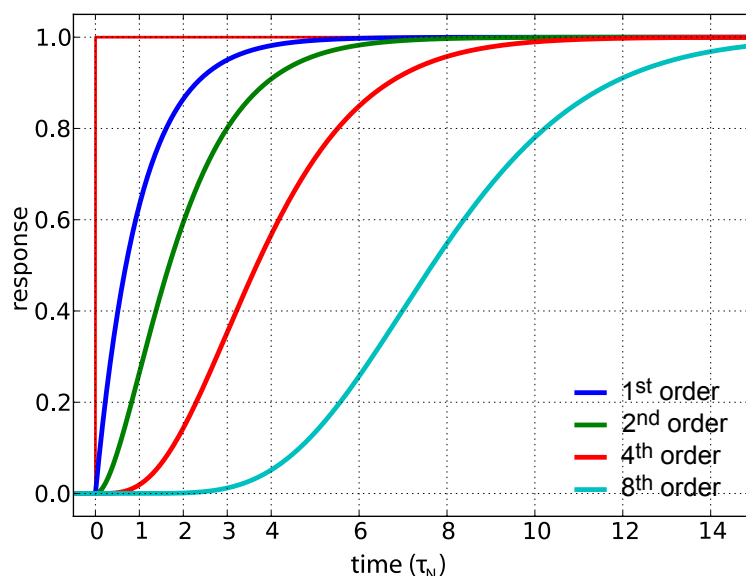


Figure 6.5. Time domain step response of the RC low-pass filters

More quantitative information on the settling time can be obtained from [Table 6.2](#). In this table, you find settling times in units of the filter time constant τ_N for all filter orders available with the MFLI Lock-in Amplifier. The values tell the time you need to wait for the filtered demodulator signal to reach 5%, 95% and 99% of the final value. This can help in making a quantitatively correct choice of filter parameters for example in a measurement involving a parameter sweep.

Table 6.2. Summary of Filter Rise Times

filter order	Setting time to		
	5%	95%	99%
1 st	$0.025 \cdot \text{TC}$	$3.0 \cdot \text{TC}$	$4.6 \cdot \text{TC}$
2 nd	$0.36 \cdot \text{TC}$	$4.7 \cdot \text{TC}$	$6.6 \cdot \text{TC}$
3 rd	$0.82 \cdot \text{TC}$	$6.3 \cdot \text{TC}$	$8.4 \cdot \text{TC}$
4 th	$1.4 \cdot \text{TC}$	$7.8 \cdot \text{TC}$	$10 \cdot \text{TC}$
5 th	$2.0 \cdot \text{TC}$	$9.2 \cdot \text{TC}$	$12 \cdot \text{TC}$
6 th	$2.6 \cdot \text{TC}$	$11 \cdot \text{TC}$	$12 \cdot \text{TC}$
7 th	$3.3 \cdot \text{TC}$	$12 \cdot \text{TC}$	$15 \cdot \text{TC}$
8 th	$4.0 \cdot \text{TC}$	$13 \cdot \text{TC}$	$16 \cdot \text{TC}$

6.4. Full Range Sensitivity

The sensitivity of the lock-in amplifier is the RMS value of an input sine that is demodulated and results in a full scale analog output. Traditionally the X, Y, or R components are mapped onto the 10 V full scale analog output. In such a case, the overall gain from input to output of the lock-in amplifier is composed of the input and output amplifier stages. Many lock-in amplifiers specify a sensitivity between 1 nV and 1 V. In other words the instrument permits an input signal between 1 nV and 1 V to be amplified to the 10 V full range output.

Analog Lock-in Amplifiers:



Digital Lock-in Amplifiers:



Figure 6.6. Sensitivity from signal input to signal output

In analog lock-in amplifiers the sensitivity is simple to understand. It is the sum of the analog amplification stages between the input and the output of the instrument: in particular the input amplifier and the output amplifier.

In digital lock-in amplifiers the sensitivity is less straightforward to understand. Analog-to-digital converters (ADC) operate with a fixed input range (e.g. 1 V) and thus require a variable-gain amplifier to amplify the input signal to the range given by the ADC. This variable-gain amplifier must be in the analog domain and its capability determines the minimum input range of the instrument. A practical analog input amplifier provides a factor 1000 amplification, thus 1 V divided by 1000 is the minimum input range of the instrument.

The input range is the maximum signal amplitude that is permitted for a given range setting. The signal is internally amplified with the suited factor, e.g. (1 mV)·1000 to result in a full swing signal at the ADC. For signals larger than the range, the ADC saturates and the signal is distorted – the measurement result becomes useless. Thus the signal should never exceed the range setting.

But the input range is not the same as the sensitivity. In digital lock-in amplifiers the sensitivity is only determined by the output amplifier, which is an entirely digital signal processing unit which performs a numerical multiplication of the demodulator output with the scaling factor. The digital output of this unit is then fed to the output digital-to-analog converter (DAC) with a fixed range of 10 V. It is this scaling factor that can be retrofitted to specify a sensitivity as known from the analog lock-in amplifiers. A large scaling factor, and thus a high sensitivity, comes at a relatively small expense for digital amplification.

One interesting aspect of digital lock-in amplifiers is the connection between input resolution and sensitivity. As the ADC operates with a finite resolution, for instance 14 bits, the minimum signal that can be detected and digitized is for instance 1 mV divided by the resolution of the ADC. With 14 bits the minimum level that can be digitized would be 122 nV. How is it possible to reach 1 nV sensitivity without using a 21 bit analog-to-digital converter? In a world without noise it is not possible. Inversely, thanks to noise and current digital technology it is possible to achieve a sensitivity even below 1 nV.

Most sources of broadband noise, including the input amplifier, can be considered as Gaussian noise sources. Gaussian noise is equally distributed in a signal, and thus generates equally

distributed disturbances. The noise itself can be filtered by the lock-in amplifier down to a level where it does not impact the measurement. Still, in the interplay with the signal, the noise does have an effect on the measurement. The input of the ADC is the sum of the noise and the signal amplitude. Every now and then, the signal amplitude on top of the large noise will be able to toggle the least significant bits even for very small signals, as low as 1 nV and below. The resulting digital signal has a component at the signal frequency and can be detected by the lock-in amplifier.

There is a similar example from biology. Rod cells in the human eye permit humans to see in very low light conditions. The sensitivity of rod cells in the human eye is as low as a single photon. This sensitivity is achieved in low light conditions by a sort of pre-charging of the cell to be sensitive to the single photon that triggers the cell to fire an impulse. In a condition with more surround light, rod cells are less sensitive and need more photons to fire.

To summarize, in digital lock-in amplifiers the full range sensitivity is only determined by the scaling factor capability of the digital output amplifier. As the scaling can be arbitrary big, 1 nV minimum full range sensitivity is achievable without a problem. Further, digital lock-in amplifiers exploit the input noise to heavily increase the sensitivity without impacting the accuracy of the measurement.

6.5. Sinc Filtering

As explained in [Section 6.1](#), the demodulated signal in an ideal lock-in amplifier has a signal component at DC and a spurious component at twice the demodulation frequency. The components at twice the demodulation frequency (called the 2ω component) is effectively removed by regular low-pass filtering. By selecting filters with small bandwidth and faster roll-offs, the 2ω component can easily be attenuated by 100 dB or more. The problem arises at low demodulation frequencies, because this forces the user to select long integration times (e.g. >60 ms for a demodulation frequency of 20 Hz) in order to achieve the same level of 2ω attenuation.

In practice, the lock-in amplifier will modulate DC offsets and non-linearities at the signal input with the demodulation frequency, resulting in a signal at the demodulation frequency (called ω component). This component is also effectively removed by the regular low-pass filters at frequencies higher than 1 kHz.

At low demodulation frequencies, and especially for applications with demodulation frequencies close to the filter bandwidth, the ω and 2ω components can affect the measurement result. Sinc filtering allows for strong attenuation of the ω and 2ω components. Technically the sinc filter is a comb filter with notches at integer multiples of the demodulation frequency (ω , 2ω , 3ω , etc.). It removes the ω component with a suppression factor of around 80 dB. The amount of 2ω component that gets removed depends on the input signal. It can vary from entirely (e.g. 80 dB) to slightly (e.g. 5 dB). This variation is not due to the sinc filter performance but depends on the bandwidth of the input signal.

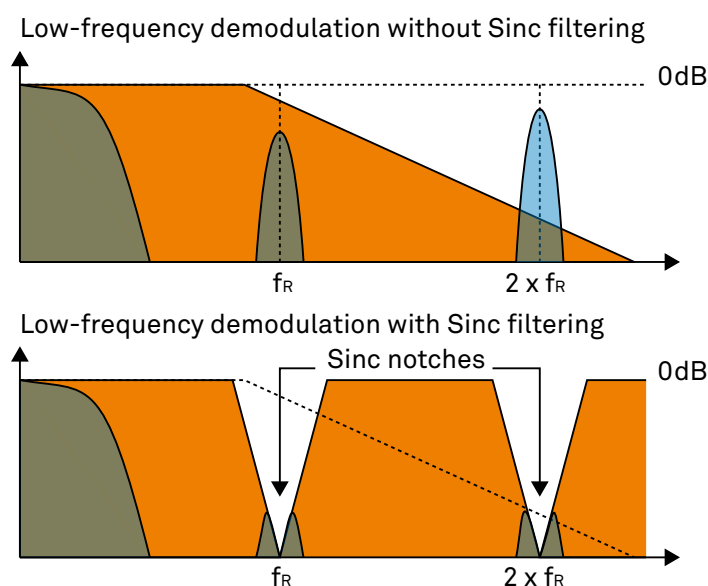


Figure 6.7. Effect of sinc filtering

Table 6.3. Artifacts in the demodulation signal

Input signal	Demodulation result before low-pass filter	Result
Signal at ω	DC component	Amplitude and phase information (wanted signal)
	2ω component	Unwanted component (can additionally be attenuated by sinc filter)

Input signal	Demodulation result before low-pass filter	Result
DC offset	ω component	Unwanted component (can additionally be attenuated by sinc filter)

We can observe the effect of the sinc filter by using the Spectrum Analyzer Tool of the MFLI Lock-in Amplifier. As an example, consider a 30 Hz signal with an amplitude of 0.1 V that demodulated using a filter bandwidth of 100 Hz and a filter order 8. In addition 0.1 V offset is added to the signal so that we get a significant ω component.

Figure 6.8 shows a spectrum with the sinc filter disabled, whereas for Figure 6.9 the sinc filter is enabled. The comparison of the two clearly shows how the sinc options dampens both the ω and 2ω components by about 100 dB.

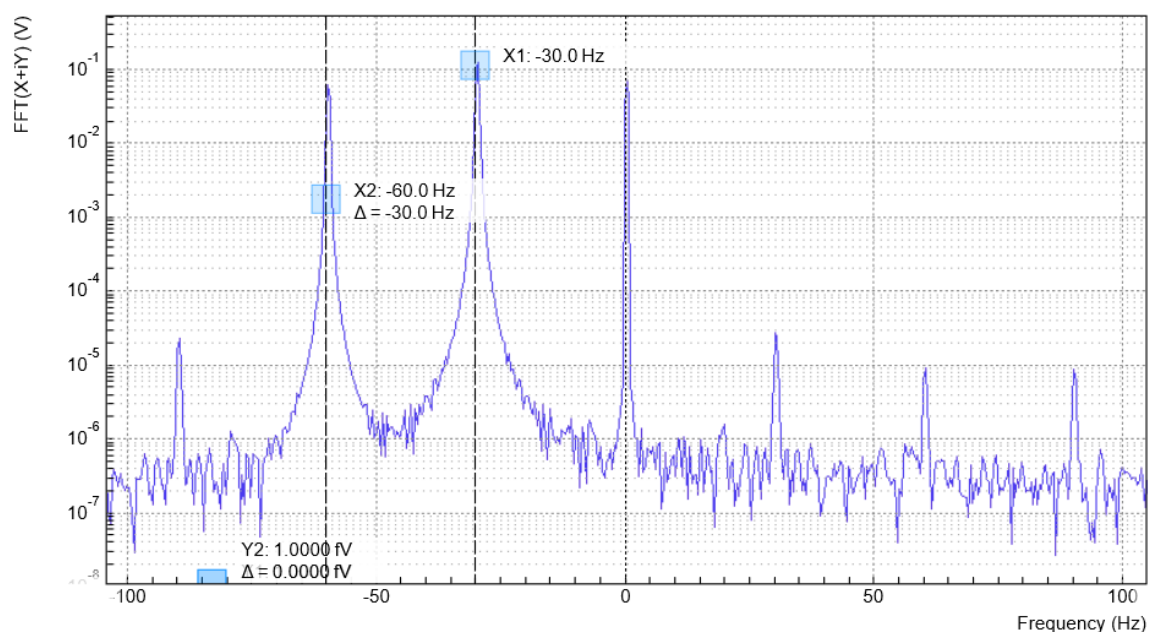


Figure 6.8. Spectrum of a demodulated 30 Hz signal without sinc filter

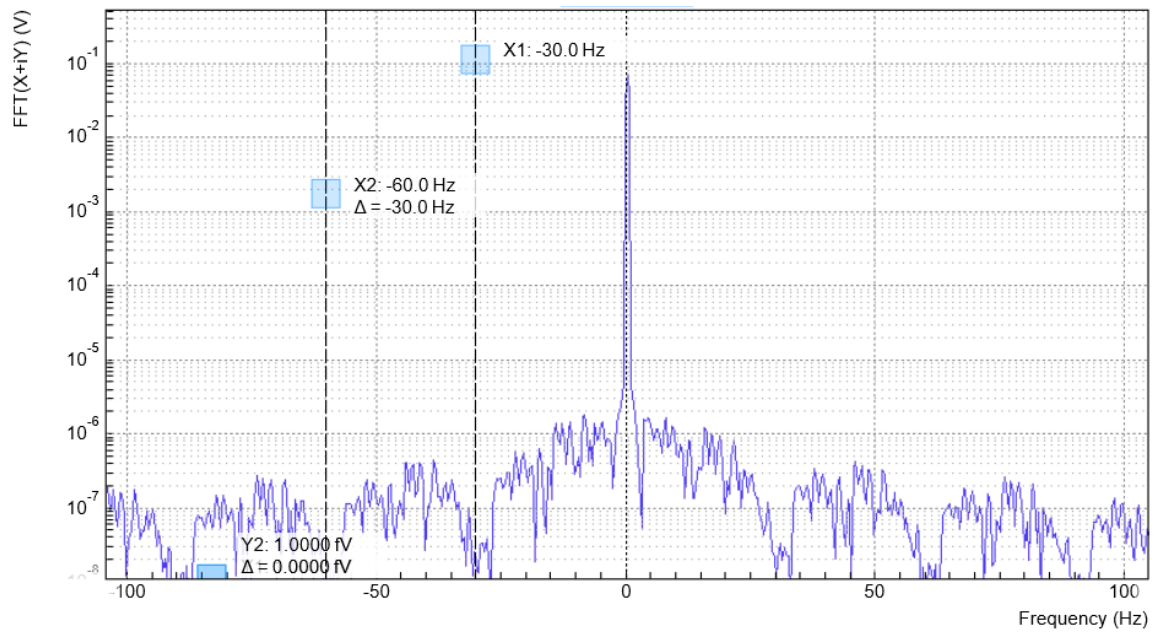


Figure 6.9. Spectrum of a demodulated 30 Hz signal with sinc filter

Note

In order to put the notches of the digital filter to ω and 2ω , the sampling rate of the filter would have to be precisely adjusted to the signal frequency. As this is technically not feasible, the generated signal frequency is adjusted instead by a very small amount.

6.6. Zoom FFT

The concept of zoom FFT allows the user to analyze the spectrum of the input signal around a particular frequency by zooming in on a narrow frequency portion of the spectrum. This is done by performing a Fourier transform of the demodulated in-phase and quadrature (X and Y) components or more precisely, on the complex quantity $X+iY$, where i is the imaginary unit. In the LabOne user interface, this functionality is available in the Spectrum tab.

In normal FFT, the sampling rate determines the frequency span and the total acquisition time determines the frequency resolution. Having a large span and a fine resolution at the same time then requires long acquisition times at high sample rates. This means that a lot of data needs to be acquired, stored, and processed, only to retain a small portion of the spectrum and discard most of it in the end. In zoom FFT, the lock-in demodulation is used to down-shift the signal frequency, thereby allowing one to use both a much lower sampling rate and sample number to achieve the same frequency resolution. Typically, to achieve a 1 Hz frequency resolution at 1 MHz, FFT would require to collect and process approximately 10^6 points, while zoom FFT only processes 10^3 points. (Of course the high rate sampling is done by the lock-in during the demodulation stage, so the zoom FFT still needs to implicitly rely on a fast ADC.)

In order to illustrate why this is so and what benefits this measurement tool brings to the user, it is useful to remind that at the end of the demodulation of the input signal $V_s(t) = A_s \cos(\omega_s t + \Theta)$, the output signal is $X + iY = F(\omega_s - \omega_r)(A_s/\sqrt{2})e^{i(\omega_s - \omega_r)t + \Theta}$ where $F(\omega)$ is the frequency response of the filters.

Since the demodulated signal has only one component at frequency $\omega_s - \omega_r$, its power spectrum (Fourier transform modulus squared) has a peak of height $(|A_s|^2/2) \cdot |F(\omega_r - \omega_s)|^2$ at $\omega_s - \omega_r$: this tells us the spectral power distribution of the input signal at frequencies close to ω_r within the demodulation bandwidth set by the filters $F(\omega)$.

Note that:

- the ability of distinguish between positive and negative frequencies works only if the Fourier transform is done on $X+iY$. Had we taken X for instance, the positive and negative frequencies of its power spectrum would be equal. The symmetry relation $G(-\omega) = G^*(\omega)$ holds for the Fourier transform $G(\omega)$ of a real function $g(t)$ and two identical peaks would appear at $\pm|\omega_s - \omega_r|$.
- one can extract the amplitude of the input signal by dividing the power spectrum by $|F(\omega)|^2$, the operation being limited by the numerical precision. This is implemented in LabOne and is activated by the Filter Compensation button: with the Filter Compensation enabled, the background noise appears white; without it, the effect of the filter roll-off becomes apparent.

The case of an input signal containing a single frequency component can be generalized to the case of multiple frequencies. In that case the power spectrum would display all the frequency components weighted by the filter transfer function, or normalized if the Filter Compensation is enabled.

When dealing with discrete-time signal processing, one has to be careful about aliasing which occurs when the signal frequencies higher than the sampling rate Ω are not sufficiently suppressed. Remember that Ω is the user settable readout rate, not the 60 MSa/s sampling rate of the MFLI input. Since the discrete-time Fourier transform extends between $-\Omega/2$ and $+\Omega/2$, the user has to make sure that at $\pm\Omega/2$ the filters provide the desired attenuation: this can be done either by increasing the sampling rate or resolving to measure a smaller frequency spectrum (i.e. with a smaller filter bandwidth).

Similarly to the continuous case, in which the acquisition time determines the maximum frequency resolution ($2\pi/T$ if T is the acquisition time), the resolution of the zoom FFT can be

increased by increasing the number of recorded data points. If N data points are collected at a sampling rate Ω , the discrete Fourier transform has a frequency resolution of Ω/N .

Chapter 7. Device Node Tree

This chapter contains reference documentation for the settings and measurement data available on MFLI Instruments. Whilst [Chapter 4](#) describes many of these settings in terms of the features available in the LabOne User Interface, this chapter describes them on the device level and provides a hierarchically organised and comprehensive list of device functionality.

Since these settings and data streams may be written and read using the LabOne APIs (Application Programming Interfaces) this chapter is of particular interest to users who would like to perform measurements programmatically via LabVIEW, Python, MATLAB, .NET or C.

Please see:

- [Section 7.1](#) for an introduction of how the instrument's settings and measurement data are organised hierarchically in the Data Server's so-called "Node Tree".
- [Section 7.2](#) for a reference list of the settings and measurement data available on MFLI Instruments, organized by branch in the Node Tree.

7.1. Introduction

This section provides an overview of how an instrument's configuration and output is organized by the Data Server.

All communication with an instrument occurs via the Data Server program the instrument is connected to (see [Section 1.5.2](#) for an overview of LabOne's software components). Although the instrument's settings are stored locally on the device, it is the Data Server's task to ensure it maintains the values of the current settings and makes these settings (and any subscribed data) available to all its current clients. A client may be the LabOne User Interface or a user's own program implemented using one of the LabOne Application Programming Interfaces, e.g., Python.

The instrument's settings and data are organized by the Data Server in a file-system-like hierarchical structure called the node tree. When an instrument is connected to a Data Server, its device ID becomes a top-level branch in the Data Server's node tree. The features of the instrument are organised as branches underneath the top-level device branch and the individual instrument settings are leaves of these branches.

For example, the auxiliary outputs of the instrument with device ID "dev2006" are located in the tree in the branch:

```
/DEV2006/AUXOUTS/
```

In turn, each individual auxiliary output channel has its own branch underneath the "AUXOUTS" branch.

```
/DEV2006/AUXOUTS/0/  
/DEV2006/AUXOUTS/1/  
/DEV2006/AUXOUTS/2/  
/DEV2006/AUXOUTS/3/
```

Whilst the auxiliary outputs and other channels are labelled on the instrument's panels and the User Interface using 1-based indexing, the Data Server's node tree uses 0-based indexing. Individual settings (and data) of an auxiliary output are available as leaves underneath the corresponding channel's branch:

```
/DEV2006/AUXOUTS/0/DEMOMODESELECT  
/DEV2006/AUXOUTS/0/LIMITLOWER  
/DEV2006/AUXOUTS/0/LIMITUPPER  
/DEV2006/AUXOUTS/0/OFFSET  
/DEV2006/AUXOUTS/0/OUTPUTSELECT  
/DEV2006/AUXOUTS/0/PREOFFSET  
/DEV2006/AUXOUTS/0/SCALE  
/DEV2006/AUXOUTS/0/VALUE
```

These are all individual node paths in the node tree; the lowest-level nodes which represent a single instrument setting or data stream. Whether the node is an instrument setting or data-stream and which type of data it contains or provides is well-defined and documented on a per-node basis in [Section 7.2](#). The different properties and types are explained in [Section 7.1.1](#).

For instrument settings, a Data Server client modifies the node's value by specifying the appropriate path and a value to the Data Server as a (path, value) pair. When an instrument's setting is changed in the LabOne User Interface, the path and the value of the node that was changed are displayed in the Status Bar in the bottom of the Window. This is described in more detail in [Section 7.1.2](#).

Module Parameters

LabOne Core Modules, such as the Sweeper, also use a similar tree-like structure to organize their parameters. Please note, however, that module nodes are not visible in the Data Server's node tree; they are local to the instance of the module created in a LabOne client and are not synchronized between clients.

7.1.1. Node Properties and Data Types

A node may have one or more of the following properties:

Read	Data can be read from the node.
Write	Data can be written to the node.
Setting	A node with write attribute corresponding to an instrument configuration. The data in these nodes will be saved to and loaded from LabOne XML settings files.
Streaming	A node with the read attribute that provides instrument data, typically at a user-configured rate. The data is usually a more complex data type, for example demodulator data is returned as <code>ZIDemodSample</code> .

A node may contain data of the following types:

Integer	Integer data.
Double	Double precision floating point data.
String	A string array.
Enumerated (integer)	As for Integer, but the node only allows certain values.
Composite data type	For example, <code>ZIDemodSample</code> . These custom data types are structures whose fields contain the instrument output, a timestamp and other relevant instrument settings such as the demodulator oscillator frequency. Documentation of custom data types is available in the C Programming chapter in the LabOne Programming Manual.

7.1.2. Exploring the Node Tree

In the LabOne User Interface

A convenient method to learn which node is responsible for a specific instrument setting is to check the Command Log history in the bottom of the LabOne User Interface. The command in the Status Bar gets updated every time a configuration change is made. [Figure 7.1](#) shows how the equivalent Matlab command is displayed after modifying the value of the auxiliary output 1's offset. The format of the LabOne UI's command history can be configured in the Config Tab (Matlab, Python and .NET are available). The entire history generated in the current UI session can be viewed by clicking the "Show Log" button.

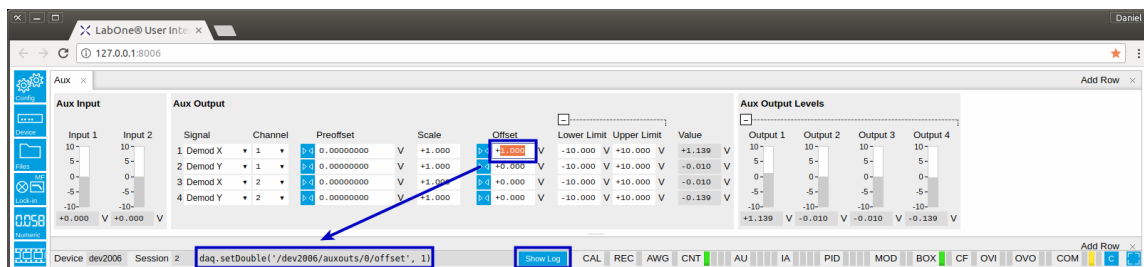


Figure 7.1. When a device's configuration is modified in the LabOne User Interface, the Status Bar displays the equivalent command to perform the same configuration via a LabOne programming interface. Here, the Matlab code to modify auxiliary output 1's offset value is provided. When "Show Log" is clicked the entire configuration history is displayed in a new browser tab.

In a LabOne Programming Interface

A list of nodes (under a specific branch) can be requested from the Data Server in an API client using the `listNodes` command (Matlab, Python, .NET) or `ziAPIListNodes()` function (C API). Please see each API's command reference for more help using the `listNodes` command. To obtain a list of all the nodes that provide data from an instrument at a high rate, so-called streaming nodes, the `streamingonly` flag can be provided to `listNodes`. More information on data streaming and streaming nodes is available in the LabOne Programming Manual).

The detailed descriptions of nodes that is provided in [Section 7.2](#) is accessible directly in the LabOne Matlab or Python programming interfaces using the "help" command. The help command is `daq.help(path)` in Python and `ziDAQ('help', path)` in Matlab. The command returns a description of the instrument node including access properties, data type, units and available options. The "help" command also handles wildcards to return a detailed description of all nodes matching the path. An example is provided below.

```
daq = zhinst.ziPython.ziDAQServer('localhost', 8004, 6)
daq.help('/dev2006/auxouts/0/offset')
# Out:
# /DEV2006/AUXOUTS/0/OFFSET
# Add the specified offset voltage to the signal after scaling. Auxiliary Output
# Value = (Signal+Preoffset)*Scale + Offset
# Properties: Read, Write, Setting
# Type: Double
# Unit: V
```

7.1.3. Data Server Nodes

The Data Server has nodes in the node tree available under the top-level `/ZI/` branch. These nodes give information about the version and state of the Data Server the client is connected to. For example, the nodes:

- `/ZI/ABOUT/VERSION`
- `/ZI/ABOUT/REVISION`

are read-only nodes that contain information about the release version and revision of the Data Server. The nodes under the `/ZI/DEVICES/` list which devices are connected, discoverable and visible to the Data Server.

The nodes:

- /ZI/CONFIG/OPEN
- /ZI/CONFIG/PORT

are settings nodes that can be used to configure which port the Data Server listens to for incoming client connections and whether it may accept connections from clients on hosts other than the localhost.

Nodes that are of particular use to programmers are:

- /ZI/DEBUG/LOGPATH - the location of the Data Server's log in the PC's filesystem,
- /ZI/DEBUG/LEVEL - the current log-level of the Data Server (configurable; has the Write attribute),
- /ZI/DEBUG/LOG - the last Data Server log entries as a string array.

For documentation of all Data Server nodes see [Section 7.2](#).

7.2. Reference Node Documentation

This section describes all the nodes in the data server's node tree organised by branch. See [Section 7.1.1](#) for an explanation of node properties and types.

7.2.1. AUXINS

/DEV..../AUXINS/n/AVERAGING

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Defines the number of samples on the input to average as a power of two. Possible values are in the range [0, 16]. A value of 0 corresponds to the sampling rate of the auxiliary input's ADC.

/DEV..../AUXINS/n/SAMPLE

Properties: Read, Stream Type: ZIAuxInSample Unit: V

Voltage measured at the Auxiliary Input after averaging. The data rate depends on the averaging value. Note, if the instrument has demodulator functionality, the auxiliary input values are available as fields in a demodulator sample and are aligned by timestamp with the demodulator output.

/DEV..../AUXINS/n/VALUES/n

Properties: Read Type: Double Unit: V

Voltage measured at the Auxiliary Input after averaging. The value of this node is updated at a low rate (50 Hz); the streaming node auxins/n/sample is updated at a high rate defined by the averaging.

7.2.2. AUXOUTS

/DEV..../AUXOUTS/n/DEMODOSELECT

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Select the channel number of the selected signal source.

/DEV..../AUXOUTS/n/LIMITLOWER

Properties: Read, Write, Setting Type: Double Unit: V

Lower limit for the signal at the Auxiliary Output. A smaller value will be clipped.

/DEV..../AUXOUTS/n/LIMITUPPER

Properties: Read, Write, Setting Type: Double Unit: V

Upper limit for the signal at the Auxiliary Output. A larger value will be clipped.

/DEV..../AUXOUTS/n/OFFSET

Properties: Read, Write, Setting Type: Double Unit: V

Add the specified offset voltage to the signal after scaling. Auxiliary Output Value = (Signal + Preoffset)*Scale + Offset

/DEV..../AUXOUTS/n/OUTPUTSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the signal source to be represented on the Auxiliary Output.

Allowed Values:

-1	"manual": Select Manual as the output option.	5	"pid": Select PID Out as the output option.
0	"demod_x": Select Demod X as the output option.	9	"pid_shift": Select PID Shift as the output option.
1	"demod_y": Select Demod Y as the output option.	10	"pid_error": Select PID Error as the output option.
2	"demod_r": Select Demod R as the output option.	11	"tu_filtered": Select TU Filtered Value as the output option.
3	"demod_theta": Select Demod Theta as the output option.	13	"tu_output": Select TU Output Value as the output option.

/DEV..../AUXOUTS/n/PREOFFSET

Properties: Read, Write, Setting Type: Double Unit: Dependent

Add a pre-offset to the signal before scaling is applied. Auxiliary Output Value = (Signal + Preoffset)*Scale + Offset

/DEV..../AUXOUTS/n/SCALE

Properties: Read, Write, Setting Type: Double Unit: None

Multiplication factor to scale the signal. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

/DEV..../AUXOUTS/n/VALUE

Properties: Read

Type: Double

Unit: V

Voltage present on the Auxiliary Output. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

7.2.3. CLOCKBASE

/DEV..../CLOCKBASE

Properties: Read

Type: Double

Unit: Hz

Returns the internal clock frequency of the device.

7.2.4. CURRINS

/DEV..../CURRINS/n/AUTORANGE

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Automatic adjustment of the Range to about two times the maximum current input amplitude measured over about 100 ms.

/DEV..../CURRINS/n/FLOAT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Switches the input between floating (ON) and connected grounds (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

Allowed Values:

0 "off": OFF: GND connected

1 "on": ON: Floating

/DEV..../CURRINS/n/MAX

Properties: Read, Write

Type: Double

Unit: A

Gives the maximum measured input current (peak value).

/DEV..../CURRINS/n/MIN

Properties: Read, Write

Type: Double

Unit: A

Gives the minimum measured input current (peak value).

/DEV..../CURRINS/n/ON

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enables the current input.

/DEV..../CURRINS/n/RANGE

Properties: Read, Write, Setting

Type: Double

Unit: A

Defines the gain of the current input amplifier. The range should exceed the incoming signal by roughly a factor of two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.

/DEV..../CURRINS/n/SCALING

Properties: Read, Write, Setting

Type: Double

Unit: None

Applies the given scale factor to the current input.

/DEV..../CURRINS/n/RANGESTEP/TRIGGER

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Switches to the next appropriate input range such that the range fits best with the measured input current.

7.2.5. DEMODS

/DEV..../DEMODS/n/ADCSELECT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the input signal for the demodulator.

Allowed Values:

0	"sigin0", "signal_input0": Sig In 1	6	"auxout2", "auxiliary_output2": Aux Out 3
1	"currin0", "current_input0": Curr In 1	7	"auxout3", "auxiliary_output3": Aux Out 4
2	"trigin0", "trigger_input0": Trigger Input 1	8	"auxin0", "auxiliary_input0": Aux In 1
3	"trigin1", "trigger_input1": Trigger Input 2		
4	"auxout0", "auxiliary_output0": Aux Out 1		

5	"auxout1", "auxiliary_output1": Aux Out 2	9	"auxin1", "auxiliary_input1": Aux In 2
174	"demod_constant_input": Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).		

/DEV..../DEMOSDS/n/BYPASS

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Allows to bypass the demodulator low-pass filter, thus increasing the bandwidth.

/DEV..../DEMOSDS/n/ENABLE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Enables the data acquisition for the corresponding demodulator. Note: increasing number of active demodulators increases load on the physical connection to the host computer.

Allowed Values:

0	"off": OFF: demodulator inactive	1	"on": ON: demodulator active
---	----------------------------------	---	------------------------------

/DEV..../DEMOSDS/n/FREQ

Properties: Read Type: Double Unit: Hz

Indicates the frequency used for demodulation and for output generation. The demodulation frequency is calculated with oscillator frequency times the harmonic factor. When the MOD option is used linear combinations of oscillator frequencies including the harmonic factors define the demodulation frequencies.

/DEV..../DEMOSDS/n/HARMONIC

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Multiplies the demodulator's reference frequency by an integer factor. If the demodulator is used as a phase detector in external reference mode (PLL), the effect is that the internal oscillator locks to the external frequency divided by the integer factor.

/DEV..../DEMOSDS/n/ORDER

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct.

Allowed Values:

1	1st order filter 6 dB/oct	5	5th order filter 30 dB/oct
2	2nd order filter 12 dB/oct	6	6th order filter 36 dB/oct
3	3rd order filter 18 dB/oct	7	7th order filter 42 dB/oct
4	4th order filter 24 dB/oct	8	8th order filter 48 dB/oct

/DEV..../DEMOSDS/n/OSCSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Connects the demodulator with the supplied oscillator. Number of available oscillators depends on the installed options.

Allowed Values:

0	Oscillator 1	2	Oscillator 3
1	Oscillator 2	3	Oscillator 4

/DEV..../DEMOSDS/n/PHASEADJUST

Properties: Read, Write Type: Integer (64 bit) Unit: None

Adjust the demodulator phase automatically in order to read 0 degrees.

/DEV..../DEMOSDS/n/PHASESHIFT

Properties: Read, Write, Setting Type: Double Unit: deg

Phase shift applied to the reference input of the demodulator.

/DEV..../DEMOSDS/n/RATE

Properties: Read, Write, Setting Type: Double Unit: 1/s

Defines the demodulator sampling rate, the number of samples that are sent to the host computer per second. A rate of about 7-10 higher as compared to the filter bandwidth usually provides

sufficient aliasing suppression. This is also the rate of data received by LabOne Data Server and saved to the computer hard disk. This setting has no impact on the sample rate on the auxiliary outputs connectors. Note: the value inserted by the user may be approximated to the nearest value supported by the instrument.

/DEV..../DEMOSDS/n/SAMPLE

Properties: Read, Stream

Type: ZIDemodSample

Unit: Dependent

Contains streamed demodulator samples with sample interval defined by the demodulator data rate.

/DEV..../DEMOSDS/n/SINC

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enables the sinc filter. When the filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. The sinc is an additional filter that attenuates these unwanted components in the demodulator output.

/DEV..../DEMOSDS/n/TIMECONSTANT

Properties: Read, Write, Setting

Type: Double

Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the demodulator low pass filter.

/DEV..../DEMOSDS/n/TRIGGER

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the acquisition mode (i.e. triggering) or the demodulator.

Allowed Values:

0	"continuous": Continuous: demodulator data is continuously streamed to the host computer.	12	"trigin1_both", "trigger_input1_both": Trigger Input 2: triggering on both rising and falling edge.
1	"trigin0_rising", "trigger_input0_rising": Trigger Input 1: rising edge triggered.	15	"trigin0or1_both", "trigger_input0or1_both": Trigger Input 1 or 2: triggering on both rising and falling edge or either trigger input.
2	"trigin0_falling", "trigger_input0_falling": Trigger Input 1: falling edge triggered.	16	"trigin0_low", "trigger_input0_low": Trigger Input 1: demodulator data is streamed to the host computer when the level is low (TTL).
3	"trigin0_both", "trigger_input0_both": Trigger Input 1: triggering on both rising and falling edge.	32	"trigin0_high", "trigger_input0_high": Trigger Input 1 for MF: demodulator
4	"trigin1_rising", "trigger_input1_rising": Trigger Input 2: rising edge triggered.		

5	"trigin0or1_both", "trigger_input0or1_both": Trigger Input 1 or 2: rising edge triggered on either input.		
8	"trigin1_falling", "trigger_input1_falling": Trigger Input 2: falling edge triggered.		
10	"trigin0or1_falling", "trigger_input0or1_falling": Trigger Input 1 or 2: falling edge triggered on either input.		
		64	"trigin1_low", "trigger_input1_low": Trigger Input 2: demodulator data is streamed to the host computer when the level is low (TTL).
		80	"trigin0or1_low", "trigger_input0or1_low": Trigger Input 1 or 2: demodulator data is streamed to the host computer when either level is low (TTL).
		128	"trigin1_high", "trigger_input1_high": Trigger Input 2: demodulator data is streamed to the host computer when the level is high (TTL).
		160	"trigin0or1_high", "trigger_input0or1_high": Trigger Input 1 or 2: demodulator data is streamed to the host computer when either level is high (TTL).

7.2.6. DIOS (Digital I/O)

```
/DEV..../DIOS/n/AUXDRIVE
```

Properties: Read Type: Integer (64 bit) Unit: None

Not used. Reserved for future use.

/DEV..../DIOS/n/DECIMATION

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Sets the decimation factor for DIO data streamed to the host computer.

```
/DEV..../DIOS/n/DRIVE
```

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

When on (1), the corresponding 8-bit bus is in output mode. When off (0), it is in input mode. Bit 0 corresponds to the least significant byte. For example, the value 1 drives the least significant byte, the value 8 drives the most significant byte.

```
/DEV..../DIOS/n/EXTCLK
```

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select DIO internal or external clocking.

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "internal": The DIO is internally clocked with a fixed frequency of 60 MHz. | 1 | "external": The DIO is externally clocked with a clock signal connected to DIO Pin 68. The available range is from 1 Hz up to the internal clock frequency. |
|---|---|---|---|

/DEV..../DIOS/n/INPUT

Properties: Read, Stream Type: ZIDIOSample Unit: None

Gives the value of the DIO input for those bytes where drive is disabled.

/DEV..../DIOS/n/MODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select DIO mode

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "manual": Manual setting of the DIO output value. | 3 | "threshold_unit": Enables setting of DIO output values by the threshold unit. |
|---|---|---|---|

/DEV..../DIOS/n/OUTPUT

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Sets the value of the DIO output for those bytes where 'drive' is enabled.

7.2.7. EXTREFS

/DEV..../EXTREFS/n/ADCSELECT

Properties: Read Type: Integer (enumerated) Unit: None

Indicates the input signal selection for the selected demodulator.

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "signin0", "signal_input0": Signal Input 1 is connected to the corresponding demodulator. | 5 | "auxout1", "auxiliary_output1": Auxiliary Output 2 is connected to the corresponding demodulator. |
| 1 | "currin0", "current_input0": Current Input 1 is connected to the corresponding demodulator. | 6 | "auxout2", "auxiliary_output2": Auxiliary Output 3 is connected to the corresponding demodulator. |
| 2 | "trigin0", "trigger_input0": Trigger Input 1 is connected to the corresponding demodulator. | 7 | "auxout3", "auxiliary_output3": Auxiliary Output 4 is connected to the corresponding demodulator. |

- | | | | |
|---|---|---|--|
| 3 | "trigin1", "trigger_input1": Trigger Input 2 is connected to the corresponding demodulator. | 8 | "auxin0", "auxiliary_input0": Auxiliary Input 1 is connected to the corresponding demodulator. |
| 4 | "auxout0", "auxiliary_output0": Auxiliary Output 1 is connected to the corresponding demodulator. | 9 | "auxin1", "auxiliary_input1": Auxiliary Input 2 is connected to the corresponding demodulator. |

/DEV..../EXTREFS/n/AUTOMODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

This defines the type of automatic adaptation of parameters in the PID used for Ext Ref.

Allowed Values:

- | | | | |
|---|--|---|--|
| 0 | "off": No automatic adaption. | 3 | "high_bandwidth": The PID coefficients, the filter bandwidth and the output limits are automatically set using a high bandwidth. |
| 1 | "coefficients_only": The coefficients of the PID controller are automatically set. | 4 | "all": All parameters of the PID including the center frequency are adapted. |
| 2 | "low_bandwidth": The PID coefficients, the filter bandwidth and the output limits are automatically set using a low bandwidth. | | |

/DEV..../EXTREFS/n/DEMODSELECT

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Indicates the demodulator connected to the extref channel.

/DEV..../EXTREFS/n/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the external reference.

/DEV..../EXTREFS/n/LOCKED

Properties: Read Type: Integer (64 bit) Unit: None

Indicates whether the external reference is locked.

/DEV..../EXTREFS/n/OSCSELECT

Properties: Read Type: Integer (64 bit) Unit: None

Indicates which oscillator is being locked to the external reference.

7.2.8. FEATURES

/DEV..../FEATURES/CODE

Properties: Read, Write Type: String Unit: None

Node providing a mechanism to write feature codes.

/DEV..../FEATURES/DEVTYPE

Properties: Read Type: String Unit: None

Returns the device type.

/DEV..../FEATURES/OPTIONS

Properties: Read Type: String Unit: None

Returns enabled options.

/DEV..../FEATURES/SERIAL

Properties: Read Type: String Unit: None

Device serial number.

7.2.9. IMPS (Impedance)

/DEV..../IMPS/n/AC

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.

/DEV..../IMPS/n/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enable impedance calculation for demodulator data.

/DEV..../IMPS/n/FREQ

Properties: Read, Write, Setting Type: Double Unit: Hz

Frequency control for the oscillator used for impedance measurement.

/DEV..../IMPS/n/IMPEDANCERANGE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select a fixed impedance range used for the measurement.

Allowed Values:

- | | | | |
|---|----------------------------|---|----------------------------|
| 0 | "100_Ohm": 100 Ω | 5 | "10_MOhm": 10 M Ω |
| 1 | "1_kOhm": 1 k Ω | 6 | "100_MOhm": 100 M Ω |
| 2 | "10_kOhm": 10 k Ω | 7 | "1_GOhm": 1 G Ω |
| 3 | "100_kOhm": 100 k Ω | 8 | "10_GOhm": 10 G Ω |
| 4 | "1_MOhm": 1 M Ω | | |

/DEV..../IMPS/n/INTERPOLATION

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the interpolation method of the compensation data. The interpolation method is particularly important if the derivative changes strongly e.g at cut-off frequencies.

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "linear": Linear: The linear interpolation is fastest but may create compensation errors in between the frequency points used for compensation. | 1 | "pchip": Piecewise Cubic Hermite (PCHIP): The piecewise cubic Hermite interpolation will result in very accurate results but requires more calculation power. |
|---|---|---|---|

/DEV..../IMPS/n/MAXBANDWIDTH

Properties: Read, Write, Setting Type: Double Unit: Hz

Limit of the maximum bandwidth used on the demodulator filter. Values above 1 kHz can heavily diminish measurement accuracy in the high-frequency region where the amplitude is no more constant over frequency.

/DEV..../IMPS/n/MODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select impedance measurement mode.

Allowed Values:

- | | | | |
|---|--|---|--|
| 0 | 4 Terminal: This method uses the current and the voltage drop across the DUT to calculate the DUT impedance. This method results in very accurate measurements as influences of series | 1 | 2 Terminal: This method uses the driving voltage and the measured current to calculate the DUT impedance. The 2 Terminal method can be beneficial when measuring very high impedance where the parasitics of the voltage |
|---|--|---|--|

resistors on the output and current input are excluded.

measurement are limiting the frequency range.

/DEV..../IMPS/n/MODEL

Properties: Read, Write, Setting

Type: Integer (enumerated) Unit: None

Representation of the complex impedance value Z by two real values accessible as Parameter 1 and Parameter 2 on all user interface displays.

Allowed Values:

- | | |
|---|--|
| <p>0 "r_c_parallel": $R_p \parallel C_p$: Impedance value Z is represented by a resistive element R_p in parallel with a capacitive element C_p.</p> | <p>5 "q_c_series": $Q C_s$: Impedance value Z is represented by a quality factor $Q = -\text{Imag}(Z)/\text{Real}(Z)$ and a series capacitive element.</p> |
| <p>1 "r_c_series": $R_s + C_s$: Impedance value Z is represented by a resistive element R_s in series with a capacitive element C_s.</p> | <p>6 "d_L_series": $D L_s$: Impedance value Z is represented by a dissipation factor $D = \text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a series inductive element.</p> |
| <p>2 "r_L_series": $R_s + L_s$: Impedance value Z is represented by a resistive element R_s in series with an inductive element L_s.</p> | <p>7 "q_L_series": $Q L_s$: Impedance value Z is represented by a quality factor $Q = \text{Imag}(Z)/\text{Real}(Z)$ and a series inductive element.</p> |
| <p>3 "g_b_parallel", "admittance": $G B$: Impedance value Z is represented by conductance $G = \text{Real}(Y)$ and Susceptance $B = \text{Imag}(Y)$ of the admittance $Y = 1/Z$.</p> | <p>8 "r_L_parallel": $R_p \parallel L_p$: Impedance value Z is represented by a resistive element R_p in parallel with an inductive element L_p.</p> |
| <p>4 "d_c_series": $D C_s$: Impedance value Z is represented by a dissipation factor $D = -\text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a series capacitive element.</p> | <p>9 "d_c_parallel": $D C_p$: Impedance value Z is represented by a dissipation factor $D = \text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a parallel capacitive element C_p.</p> |

/DEV..../IMPS/n/OMEGASUPPRESSION

Properties: Read, Write, Setting

Type: Double

Unit: dB

Suppression of the omega and 2-omega components. Small omega suppression can diminish measurements of very low or high impedance because the DC component can become dominant. Large omega suppression will have a significant impact on sweep time especially for low filter orders.

/DEV..../IMPS/n/SAMPLE

Properties: Read, Stream

Type: ZIImpedanceSample

Unit: None

Streaming node containing the impedance measurement sample data.

/DEV..../IMPS/n/AUTO/BW

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enable automatic bandwidth control. If enabled the optimum bandwidth is calculated based on the frequency and measurement data.

/DEV..../IMPS/n/AUTO/INPUTRANGE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select input range control mode.

Allowed Values:

- | | |
|--|---|
| 0 Manual: In manual mode the current and voltage input ranges are adjusted manually and separately. Use this mode with care as overload will result in inaccurate impedance results. | 2 Impedance range is fixed. The current and voltage inputs are adjusted so that the selected impedance can be measured with the currently active excitation with best precision. Use this mode to avoid automatic range adjustments in a frequency sweep. |
| 1 Auto: Dynamically adjust the input range according to the measured input signal strength. This optimizes the dynamic range and precision of impedance measurements. | |

/DEV..../IMPS/n/AUTO/OUTPUT

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

If enabled, the drive voltage amplitude is controlled by the device. If disabled it can be set manually.

/DEV..../IMPS/n/AUTO/SUPPRESS

Properties: Read, Write Type: Integer (64 bit) Unit: None

Indicates disabled periodic auto range control. A running sweeper module takes over the range control and thus disables the periodic range checks.

/DEV..../IMPS/n/BIAS/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the DC bias voltage.

/DEV..../IMPS/n/BIAS/VALUE

Properties: Read, Write, Setting Type: Double Unit: V

DC bias voltage applied across the device under test. Both positive and negative bias voltages are supported. In a 4-terminal measurement, the bias voltage is limited by the maximum common voltage input range of the device. In a 2-terminal measurement, the bias voltage can be larger because the voltage inputs are not connected.

/DEV..../IMPS/n/CURRENT/DEMOMDSELECT

Properties: Read Type: Integer (64 bit) Unit: None

Demodulator used for current demodulation.

/DEV..../IMPS/n/CURRENT/INPUTSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the current input used for two- and four-terminal impedance measurements.

Allowed Values:

0	"currin0", "current_input0": Current Input 1	9	"auxin1", "auxiliary_input1": Aux In 2
---	--	---	--

```
8  "auxin0", "auxiliary_input0": Aux In 1
```

```
/DEV..../IMPS/n/CURRENT/INVERT
```

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

If enabled, the current input signal is inverted. This is useful to switch the polarity of an input signal which can be caused by additional current amplifiers.

```
/DEV..../IMPS/n/CURRENT/RANGE
```

Properties: Read, Write, Setting Type: Double Unit: A

Input current range used for the impedance measurement. Small current input ranges have a reduced bandwidth. In the Range Control modes 'Auto' and 'Impedance', the current range is switched automatically to a higher range if the frequency is too high.

/DEV..../IMPS/n/CURRENT/SCALING

Properties: Read, Write, Setting Type: Double Unit: A/V

The scaling factor will be applied to the current measurement done with an Aux In input.

/DEV..../IMPS/n/DEMOD/HARMONIC

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Multiplies the demodulator's reference frequency with the integer factor defined by this field.

```
/DEV..../IMPS/n/DEMODO/ORDER
```

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct of the current demodulator.

Allowed Values:

1	1st order filter 6 dB/oct	5	5th order filter 30 dB/oct
2	2nd order filter 12 dB/oct	6	6th order filter 36 dB/oct
3	3rd order filter 18 dB/oct	7	7th order filter 42 dB/oct
4	4th order filter 24 dB/oct	8	8th order filter 48 dB/oct

/DEV..../IMPS/n/DEMODO/OSCSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Oscillator used to generate the frequency of the excitation voltage on the Hcur (+V) connector.

Allowed Values:

0	Oscillator 1	2	Oscillator 3
1	Oscillator 2	3	Oscillator 4

/DEV..../IMPS/n/DEMODO/RATE

Properties: Read, Write, Setting Type: Double Unit: 1/s

Impedance data streaming rate. The same data rate is applied to the demodulators that are used for the impedance measurement.

/DEV..../IMPS/n/DEMODO/SINC

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the sinc filter.

/DEV..../IMPS/n/DEMODO/TIMECONSTANT

Properties: Read, Write, Setting Type: Double Unit: s

Defines low pass filter time constant.

/DEV..../IMPS/n/DISCARD/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Discarding impedance samples outside the indicated range.

/DEV..../IMPS/n/DISCARD/LIMITLOWER

Properties: Read, Write, Setting Type: Double Unit: None

Threshold for $\text{abs}(Z)$ below which the impedance samples are discarded.

/DEV..../IMPS/n/DISCARD/LIMITUPPER

Properties: Read, Write, Setting Type: Double Unit: None

Threshold for $\text{abs}(Z)$ above which the impedance samples are discarded.

/DEV..../IMPS/n/ONEPERIOD/ACTIVE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables one-period averaging for low frequency impedance measurements. The LED is green when active.

/DEV..../IMPS/n/ONEPERIOD/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables one-period averaging for low frequency impedance measurements. The LED is green when active.

/DEV..../IMPS/n/OUTPUT/AMPLITUDE

Properties: Read, Write, Setting Type: Double Unit: V

Drive amplitude on the Signal Output.

/DEV..../IMPS/n/OUTPUT/DEMOMD

Properties: Read Type: Integer (64 bit) Unit: None

Demodulator unit used to generate the excitation voltage on the Signal Output.

/DEV..../IMPS/n/OUTPUT/ON

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Main switch for the Signal Output corresponding to the blue LED indicator on the instrument front panel.

/DEV..../IMPS/n/OUTPUT/RANGE

Properties: Read, Write, Setting Type: Double Unit: V

Selects the output voltage range.

Properties: Read Type: Integer (64 bit) Unit: None

Indicates if the internal calibration is applied to the measurement data.

/DEV..../IMPS/n/CALIB/INTERNAL/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the internal calibration. This ensures that the input range gains match over the full frequency range. With enabled internal calibration the device fulfills the impedance accuracy specification. The internal calibration is a prerequisite to apply a user compensation.

/DEV..../IMPS/n/CALIB/INTERNAL/SMOOTH

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables smoothing of the internal calibration data. This results in lower noise in the measured data.

/DEV..../IMPS/n/CALIB/USER/ACTIVE

Properties: Read Type: Integer (64 bit) Unit: None

Indicates that a valid user compensation is active. If active the impedance data streams deliver amplitude and phase corrected data based on the impedance user compensation.

/DEV..../IMPS/n/CALIB/USER/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the user compensation of the impedance data. The user compensation is correcting parasitics and delays caused by the external setup. The user compensation is applied on top of the internal impedance calibration.

/DEV..../IMPS/n/CALIB/USER/SMOOTH

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables smoothing of the compensation data. This results in lower noise in the measured data.

/DEV..../IMPS/n/CONFIDENCE/COMPENSATION/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the indication of strong compensation in the plots. A strong compensation diminishes the measurement accuracy of the parameter.

/DEV..../IMPS/n/CONFIDENCE/COMPENSATION/RATIO

Properties: Read, Write, Setting Type: Double Unit: None

Strength of the compensation that will trigger the strong compensation warning.

/DEV..../IMPS/n/CONFIDENCE/FREQLIMIT/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the frequency limit detection based on the used current input range. Only relevant when Range Control is set to Manual.

/DEV..../IMPS/n/CONFIDENCE/LOWDUT2T/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables a warning when measuring a low impedance (100k) with a 2 point contact.

/DEV..../IMPS/n/CONFIDENCE/LOWDUT2T/RATIO

Properties: Read, Write, Setting Type: Double Unit: None

When measuring with 2 point contacts, too low impedance of DUT will trigger the Low DUT 2T warning.

/DEV..../IMPS/n/CONFIDENCE/ONEPERIOD/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the detection of unreliable data points where data sample loss leads to an invalid one-period average. Try reducing the data transfer rate.

/DEV..../IMPS/n/CONFIDENCE/OPENDETECT/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the open terminal detection for 4-terminal measurements.

/DEV..../IMPS/n/CONFIDENCE/OPENDETECT/RATIO

Properties: Read, Write, Setting Type: Double Unit: None

Open terminal detection ratio. An open terminal is reported if the excitation calculated from current and voltage drop differs more than the specified factor from the driving voltage.

/DEV..../IMPS/n/CONFIDENCE/OVERFLOW/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the overload detection for current and voltage.

/DEV..../IMPS/n/CONFIDENCE/QFACTOR/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the detection of negative Q or D factors. Negative Q or D factors mean the measured impedance does not correspond to the chosen Representation. This can be due to an erroneous compensation, a bad choice of the Representation, or noise.

/DEV..../IMPS/n/CONFIDENCE/SUPPRESSION/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

The Suppression Confidence Indicator indicates if one of the two parameters of a circuit representation cannot be calculated reliably from the measured impedance. This is the case if a small variation in one (dominant) representation parameter creates a strong variation of the other (suppressed) representation parameter. Such an error amplification indicates that the measurement of the secondary parameter is unreliable.

/DEV..../IMPS/n/CONFIDENCE/SUPPRESSION/RATIO

Properties: Read, Write, Setting Type: Double Unit: None

Error amplification limit for which a secondary parameter is marked unreliable. Larger gain values mean larger warning tolerances. A gain value between 10 and 100 is best.

/DEV..../IMPS/n/CONFIDENCE/UNDERFLOW/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the underflow detection for current and voltage.

/DEV..../IMPS/n/CONFIDENCE/UNDERFLOW/RATIO

Properties: Read, Write, Setting Type: Double Unit: None

The underflow condition is met if the measured amplitude is lower than the specified ratio relative to full scale.

/DEV..../IMPS/n/CURRENT/PID/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

If enabled, the PID value is used for the impedance calculation instead of the measured value at the current input.

7.2.10. MODS (Modulation)

/DEV..../MODS/n/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the modulation.

/DEV..../MODS/n/FREQDEV

Properties: Read, Write, Setting Type: Double Unit: Hz

FM mode peak deviation value.

/DEV..../MODS/n/FREQDEVENABLE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

In FM mode, choose to work with either modulation index or peak deviation. The modulation index equals peak deviation divided by modulation frequency.

Allowed Values:

0 "modulation_index": Use modulation index. 1 "peak_deviation": Use peak deviation.

/DEV..../MODS/n/INDEX

Properties: Read, Write, Setting Type: Double Unit: None

FM modulation index: The modulation index equals peak deviation divided by modulation frequency.

/DEV..../MODS/n/MODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the modulation mode.

Allowed Values:

0 "am": AM Modulation 2 "manual": Manual
1 "fm": FM Modulation

/DEV..../MODS/n/OUTPUT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select Signal Output.

Allowed Values:

0 "none": None

1 "sigout0", "signal_output0": Signal Output 1

/DEV..../MODS/n/CARRIER/AMPLITUDE

Properties: Read, Write, Setting

Type: Double

Unit: V

Set the carrier amplitude

/DEV..../MODS/n/CARRIER/ENABLE

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enable the modulation

/DEV..../MODS/n/CARRIER/HARMONIC

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Set the harmonic of the carrier frequency. 1 = Fundamental

/DEV..../MODS/n/CARRIER/INPUTSELECT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select Signal Input for the carrier demodulation

Allowed Values:

0 "sigin0", "signal_input0": Sig In 1

1 "currin0", "current_input0": Curr In 1

2 "trigin0", "trigger_input0": Trigger Input 1

3 "trigin1", "trigger_input1": Trigger Input 2

4 "auxout0", "auxiliary_output0": Aux Out 1

5 "auxout1", "auxiliary_output1": Aux Out 2

6 "auxout2", "auxiliary_output2": Aux Out 3

7 "auxout3", "auxiliary_output3": Aux Out 4

8 "auxin0", "auxiliary_input0": Aux In 1

9 "auxin1", "auxiliary_input1": Aux In 2

174 "demod_constant_input":
Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is

written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).

/DEV..../MODS/n/CARRIER/ORDER

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Filter order used for carrier demodulation

Allowed Values:

1	1st order filter 6 dB/oct	5	5th order filter 30 dB/oct
2	2nd order filter 12 dB/oct	6	6th order filter 36 dB/oct
3	3rd order filter 18 dB/oct	7	7th order filter 42 dB/oct
4	4th order filter 24 dB/oct	8	8th order filter 48 dB/oct

/DEV..../MODS/n/CARRIER/OSCSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the oscillator for the carrier signal.

Allowed Values:

0	Oscillator 1	2	Oscillator 3
1	Oscillator 2	3	Oscillator 4

/DEV..../MODS/n/CARRIER/PHASESHIFT

Properties: Read, Write, Setting Type: Double Unit: degree

Phase shift applied to the reference input of the carrier demodulator and also to the carrier signal on the Signal Outputs

/DEV..../MODS/n/CARRIER/TIMECONSTANT

Properties: Read, Write, Setting Type: Double Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the low-pass filter for the carrier demodulation.

/DEV..../MODS/n/SIDEBANDS/n/AMPLITUDE

Properties: Read, Write, Setting Type: Double Unit: V

Set the amplitude of the sideband components.

/DEV..../MODS/n/SIDEBANDS/n/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enable the signal generation for the respective sideband

/DEV..../MODS/n/SIDEBANDS/n/HARMONIC

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Set harmonic of the sideband frequencies. 1 = fundamental

/DEV..../MODS/n/SIDEBANDS/n/INPUTSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select Signal Input for the sideband demodulation

Allowed Values:

0	"sigin0", "signal_input0": Sig In 1	6	"auxout2", "auxiliary_output2": Aux Out 3
1	"currin0", "current_input0": Curr In 1	7	"auxout3", "auxiliary_output3": Aux Out 4
2	"trigin0", "trigger_input0": Trigger Input 1	8	"auxin0", "auxiliary_input0": Aux In 1
3	"trigin1", "trigger_input1": Trigger Input 2	9	"auxin1", "auxiliary_input1": Aux In 2
4	"auxout0", "auxiliary_output0": Aux Out 1		
5	"auxout1", "auxiliary_output1": Aux Out 2	174	"demod_constant_input": Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).

/DEV..../MODS/n/SIDEBANDS/n/MODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Enabling of the first sideband and selection of the position of the sideband relative to the carrier frequency for manual mode.

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "off": Off: First sideband is disabled. The sideband demodulator behaves like a normal demodulator. | 2 | "lower": C - M: First sideband to the left of the carrier |
| 1 | "upper": C + M: First sideband to the right of the carrier | | |

/DEV..../MODS/n/SIDEBANDS/n/ORDER

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Filter order used for sideband demodulation

Allowed Values:

- | | | | |
|---|----------------------------|---|----------------------------|
| 1 | 1st order filter 6 dB/oct | 5 | 5th order filter 30 dB/oct |
| 2 | 2nd order filter 12 dB/oct | 6 | 6th order filter 36 dB/oct |
| 3 | 3rd order filter 18 dB/oct | 7 | 7th order filter 42 dB/oct |
| 4 | 4th order filter 24 dB/oct | 8 | 8th order filter 48 dB/oct |

/DEV..../MODS/n/SIDEBANDS/n/OSCSELECT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select the oscillator for the second sideband.

Allowed Values:

- | | | | |
|---|--------------|---|--------------|
| 0 | Oscillator 1 | 2 | Oscillator 3 |
| 1 | Oscillator 2 | 3 | Oscillator 4 |

/DEV..../MODS/n/SIDEBANDS/n/PHASESHIFT

Properties: Read, Write, Setting Type: Double Unit: degree

Phase shift applied to the reference input of the sideband demodulator and also to the sideband signal on the Signal Outputs

/DEV..../MODS/n/SIDEBANDS/n/TIMECONSTANT

Properties: Read, Write, Setting Type: Double Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the low-pass filter for the sideband demodulation.

7.2.11. OSCS (Oscillators)

/DEV..../OSCS/n/FREQ

Properties: Read, Write, Setting

Type: Double

Unit: Hz

Frequency control for each oscillator.

7.2.12. PIDS

/DEV..../PIDS/n/CENTER

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

Sets the center value for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/DEV..../PIDS/n/D

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

PID derivative gain.

/DEV..../PIDS/n/DLIMITTIMECONSTANT

Properties: Read, Write, Setting

Type: Double

Unit: s

The cutoff of the low-pass filter for the D limitation given as time constant. When set to 0, the low-pass filter is disabled.

/DEV..../PIDS/n/ENABLE

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enable the PID controller.

/DEV..../PIDS/n/ERROR

Properties: Read

Type: Double

Unit: Dependent

Error = Set point - PID Input.

/DEV..../PIDS/n/I

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

PID integral gain I.

/DEV..../PIDS/n/INPUT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select input source of PID controller.

Allowed Values:

0 "demod_x": Demodulator X

3 "demod_theta": Demodulator Theta

1 "demod_y": Demodulator Y

4 "auxin", "auxiliary_input": Aux Input

2 "demod_r": Demodulator R

5 "auxout", "auxiliary_output": Aux Output

/DEV..../PIDS/n/INPUTCHANNEL

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Select input channel of PID controller.

/DEV..../PIDS/n/KEEPINT

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: Dependent

If enabled, the accumulated integral error is maintained upon restart of the PID. If is disabled, the integral error is set to zero when the PID is disabled.

/DEV..../PIDS/n/LIMITLOWER

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

Sets the lower limit for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/DEV..../PIDS/n/LIMITUPPER

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

Sets the upper limit for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/DEV..../PIDS/n/MODE

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Sets the operation mode of the PID module.

Allowed Values:

- | | | | |
|---|--------------------------------|---|---------------------------------------|
| 0 | "pid": PID | 2 | "extref": ExtRef (external reference) |
| 1 | "pll": PLL (phase locked loop) | | |

/DEV..../PIDS/n/OUTPUT

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Select output of the PID controller.

Allowed Values:

- | | | | |
|---|--|---|---|
| 0 | "sigout", "signal_output": Feedback to the main signal output amplitudes | 3 | "auxout_offset",
"auxiliary_output_offset": Feedback to any of the 4 Auxiliary Output's Offset |
| 1 | "oscillator_frequency": Feedback to any of the internal oscillator frequencies | 4 | "sigout_offset", "signal_output_offset": Feedback to the main Signal Output offset adjustment |
| 2 | "demod_phase": Feedback to any of the 8 demodulator phase set points | | |

/DEV..../PIDS/n/OUTPUTCHANNEL

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Select the output channel of the driven output of PID controller.

/DEV..../PIDS/n/P

Properties: Read, Write, Setting Type: Double Unit: Dependent

PID Proportional gain P.

/DEV..../PIDS/n/PHASEUNWRAP

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the phase unwrapping to track phase errors past the +/-180 degree boundary and increase PLL bandwidth.

/DEV..../PIDS/n/RATE

Properties: Read, Write, Setting Type: Double Unit: 1/s

PID sampling rate and update rate of PID outputs. Needs to be set substantially higher than the targeted loop filter bandwidth.

/DEV..../PIDS/n/SETPOINT

Properties: Read, Write, Setting Type: Double Unit: Dependent

PID controller setpoint

/DEV..../PIDS/n/SHIFT

Properties: Read

Type: Double

Unit: Dependent

Difference between the current output value Out and the Center. Shift = $P \cdot \text{Error} + I \cdot \text{Int}(\text{Error}, dt) + D \cdot d\text{Error}/dt$

/DEV..../PIDS/n/VALUE

Properties: Read

Type: Double

Unit: Dependent

Gives the current PID output value.

/DEV..../PIDS/n/DEMODO/ADCSELECT

Properties: Read

Type: Integer (enumerated)

Unit: None

Indicates the signal source which is connected to the chosen input demodulator channel.

Allowed Values:

- | | |
|---|---|
| 0 "sigin0", "signal_input0": Signal Input 1 is connected to the corresponding demodulator. | 5 "auxout1", "auxiliary_output1": Auxiliary Output 2 is connected to the corresponding demodulator. |
| 1 "currin0", "current_input0": Current Input 1 is connected to the corresponding demodulator. | 6 "auxout2", "auxiliary_output2": Auxiliary Output 3 is connected to the corresponding demodulator. |
| 2 "trigin0", "trigger_input0": Trigger Input 1 is connected to the corresponding demodulator. | 7 "auxout3", "auxiliary_output3": Auxiliary Output 4 is connected to the corresponding demodulator. |
| 3 "trigin1", "trigger_input1": Trigger Input 2 is connected to the corresponding demodulator. | 8 "auxin0", "auxiliary_input0": Auxiliary Input 1 is connected to the corresponding demodulator. |
| 4 "auxout0", "auxiliary_output0": Auxiliary Output 1 is connected to the corresponding demodulator. | 9 "auxin1", "auxiliary_input1": Auxiliary Input 2 is connected to the corresponding demodulator. |

/DEV..../PIDS/n/DEMODO/HARMONIC

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Multiplier of the for the reference frequency of the current demodulator.

/DEV..../PIDS/n/DEMODO/ORDER

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct of the current demodulator.

Allowed Values:

1	1st order filter 6 dB/oct	5	5th order filter 30 dB/oct
2	2nd order filter 12 dB/oct	6	6th order filter 36 dB/oct
3	3rd order filter 18 dB/oct	7	7th order filter 42 dB/oct
4	4th order filter 24 dB/oct	8	8th order filter 48 dB/oct

/DEV..../PIDS/n/DEMODO/TIMECONSTANT

Properties: Read, Write, Setting Type: Double Unit: s

Defines the characteristic time constant (cut off) of the demodulator filter used as an input.

/DEV..../PIDS/n/PLL/AUTOMODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

This defines the type of automatic adaptation of parameters in the PID.

Allowed Values:

0	"no_adaption": No automatic adaption.	3	"pid_coeffs_filter_high_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a high bandwidth.
1	"pid_coeffs": The coefficients of the PID controller are automatically set.	4	"pid_all": All parameters of the PID including the center frequency are adapted.
2	"pid_coeffs_filter_low_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a low bandwidth.		

/DEV..../PIDS/n/PLL/LOCKED

Properties: Read Type: Integer (64 bit) Unit: None

Indicates when the PID, configured as PLL, is locked.

/DEV..../PIDS/n/SETPOINTTOGGLE/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the setpoint toggle.

/DEV..../PIDS/n/SETPOINTTOGGLE/RATE

Properties: Read, Write, Setting Type: Double Unit: Hz

Defines the rate of setpoint toggling. Note that possible values are logarithmically spaced with a factor of 4 between values.

/DEV..../PIDS/n/SETPOINTTOGGLE/SETPOINT

Properties: Read, Write, Setting Type: Double Unit: Dependent

Defines the setpoint value used for setpoint toggle.

/DEV..../PIDS/n/STREAM/EFFECTIVERATE

Properties: Read Type: Double Unit: 1/s

Current rate of the PID stream data sent to PC. Defined based on Max Rate.

/DEV..../PIDS/n/STREAM/ERROR

Properties: Read, Stream Type: Double Unit: Dependent

PID Error = Set point - PID Input.

/DEV..../PIDS/n/STREAM/OVERFLOW

Properties: Read Type: Integer (64 bit) Unit: None

Indicates the streaming fifo overflow state. 0 = OK, 1 = overflow.

/DEV..../PIDS/n/STREAM/RATE

Properties: Read, Write, Setting Type: Double Unit: 1/s

Target Rate for PID output data sent to PC. This value defines the applied decimation for sending data to the PC. It does not affect any other place where PID data are used.

/DEV..../PIDS/n/STREAM/SHIFT

Properties: Read, Stream Type: Double Unit: Dependent

Gives the difference between the current output value and the center value. $\text{Shift} = P \cdot \text{Error} + I \cdot \text{Int}(\text{Error}, dt) + D \cdot d\text{Error}/dt$

/DEV..../PIDS/n/STREAM/VALUE

Properties: Read, Stream Type: Double Unit: Dependent

Gives the current PID output value.

7.2.13. SCOPES

/DEV..../SCOPES/n/CHANNEL

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Activates the scope channels

Allowed Values:

- | | | | |
|---|---------------------------|---|---|
| 1 | Only channel 1 is active. | 3 | "both": Both, channel 1 and 2 are active. |
| 2 | Only channel 2 is active. | | |

/DEV..../SCOPES/n/ENABLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the acquisition of scope shots.

/DEV..../SCOPES/n/LENGTH

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Defines the length of the recorded scope shot in number of samples.

/DEV..../SCOPES/n/SINGLE

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Puts the scope into single shot mode.

/DEV..../SCOPES/n/TIME

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Defines the time base of the scope from the divider exponent of the instrument's clock base. The resulting sampling time is $2^n/\text{clockbase}$.

Allowed Values:

- | | | | |
|---|----------|----|----------|
| 0 | 60 MHz | 8 | 234 kHz |
| 1 | 30 MHz | 9 | 117 kHz |
| 2 | 15 MHz | 10 | 58.6 kHz |
| 3 | 7.5 MHz | 11 | 29.3 kHz |
| 4 | 3.75 MHz | 12 | 14.6 kHz |
| 5 | 1.88 MHz | 13 | 7.32 kHz |
| 6 | 938 kHz | 14 | 3.66 kHz |
| 7 | 469 kHz | 15 | 1.83 kHz |

/DEV..../SCOPES/n/TRIGCHANNEL

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the trigger source signal.

Allowed Values:

0	"signin", "signal_input0": Signal Input 1	49	"demod1_r": Demod 2 R. Requires an installed digitizer (DIG) option.
1	"currin0", "current_input0": Current Input 1	50	"demod2_r": Demod 3 R. Requires an installed digitizer (DIG) option.
2	"trigin0", "trigger_input0": Trigger Input 1	51	"demod3_r": Demod 4 R. Requires an installed digitizer (DIG) option.
3	"trigin1", "trigger_input1": Trigger Input 2	64	"demod0_theta": Demod 1 Θ . Requires an installed digitizer (DIG) option.
4	"auxout0", "auxiliary_output0": Aux Output 1. Requires an installed digitizer (DIG) option.	65	"demod1_theta": Demod 2 Θ . Requires an installed digitizer (DIG) option.
5	"auxout1", "auxiliary_output1": Aux Output 2. Requires an installed digitizer (DIG) option.	66	"demod2_theta": Demod 3 Θ . Requires an installed digitizer (DIG) option.
6	"auxout2", "auxiliary_output2": Aux Output 3. Requires an installed digitizer (DIG) option.	67	"demod3_theta": Demod 4 Θ . Requires an installed digitizer (DIG) option.
7	"auxout3", "auxiliary_output3": Aux Output 4. Requires an installed digitizer (DIG) option.	80	"pid0_value": PID 1 value. Requires an installed digitizer (DIG) option.
8	"auxin0", "auxiliary_input0": Aux Input 1	81	"pid1_value": PID 2 value. Requires an installed digitizer (DIG) option.
9	"auxin1", "auxiliary_input1": Aux Input 2	82	"pid2_value": PID 3 value. Requires an installed digitizer (DIG) option.
10	"demod1": Osc Φ Demod 2	83	"pid3_value": PID 4 value. Requires an installed digitizer (DIG) option.
11	"demod3": Osc Φ Demod 4	144	"pid0_shift": PID 1 Shift. Requires an installed digitizer (DIG) option.
14	"trigout0", "trigger_output0": Trigger Output 1	145	"pid1_shift": PID 2 Shift. Requires an installed digitizer (DIG) option.
15	"trigout1", "trigger_output1": Trigger Output 2	146	"pid2_shift": PID 3 Shift. Requires an installed digitizer (DIG) option.
16	"demod0_x": Demod 1 X. Requires an installed digitizer (DIG) option.	147	"pid3_shift": PID 4 Shift. Requires an installed digitizer (DIG) option.
17	"demod1_x": Demod 2 X. Requires an installed digitizer (DIG) option.	208	"pid0_error": PID 1 Error. Requires an installed digitizer (DIG) option.
18	"demod2_x": Demod 3 X. Requires an installed digitizer (DIG) option.	209	"pid1_error": PID 2 Error. Requires an installed digitizer (DIG) option.
19	"demod3_x": Demod 4 X. Requires an installed digitizer (DIG) option.	210	"pid2_error": PID 3 Error. Requires an installed digitizer (DIG) option.
32	"demod0_y": Demod 1 Y. Requires an installed digitizer (DIG) option.		
33	"demod1_y": Demod 2 Y. Requires an installed digitizer (DIG) option.		

34	"demod2_y": Demod 3 Y. Requires an installed digitizer (DIG) option.	211	"pid3_error": PID 4 Error. Requires an installed digitizer (DIG) option.
35	"demod3_y": Demod 4 Y. Requires an installed digitizer (DIG) option.	224	"tu0_filtered": TU 1 Filtered Value. Requires an installed digitizer (DIG) option.
48	"demod0_r": Demod 1 R. Requires an installed digitizer (DIG) option.	225	"tu1_filtered": TU 2 Filtered Value. Requires an installed digitizer (DIG) option.
		226	"tu2_filtered": TU 3 Filtered Value. Requires an installed digitizer (DIG) option.
		227	"tu3_filtered": TU 4 Filtered Value. Requires an installed digitizer (DIG) option.

/DEV..../SCOPES/n/TRIGDELAY

Properties: Read, Write, Setting Type: Double Unit: s

Trigger position relative to reference. A positive delay results in less data being acquired before the trigger point, a negative delay results in more data being acquired before the trigger point.

/DEV..../SCOPES/n/TRIGENABLE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

When triggering is enabled scope data are acquired every time the defined trigger condition is met.

Allowed Values:

0	"on": ON: Trigger based scope shot acquisition	1	"off": OFF: Continuous scope shot acquisition
---	--	---	---

/DEV..../SCOPES/n/TRIGFALLING

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

When set (1), enables falling edge triggering. This settings is synchronized with the settings done in the /TRIGSLOPE node.

/DEV..../SCOPES/n/TRIGFORCE

Properties: Read, Write Type: Integer (64 bit) Unit: None

Forces a trigger event.

/DEV..../SCOPES/n/TRIGHOLDOFF

Properties: Read, Write, Setting Type: Double Unit: s

Defines the time before the trigger is rearmed after a recording event.

/DEV..../SCOPES/n/TRIGHOLDOFFCOUNT

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Defines the trigger event number that will trigger the next recording after a recording event. A value of '1' will start a recording for each trigger event.

/DEV..../SCOPES/n/TRIGHOLDOFFMODE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Selects the holdoff mode.

Allowed Values:

- | | | | |
|---|---|---|---|
| 0 | "time": Holdoff is defined as time (s). | 1 | "number_of_events": Holdoff is defined as number of events. |
|---|---|---|---|

/DEV..../SCOPES/n/TRIGLEVEL

Properties: Read, Write, Setting Type: Double Unit: V

Defines the trigger level.

/DEV..../SCOPES/n/TRIGREFERENCE

Properties: Read, Write, Setting Type: Double Unit: None

Trigger reference position relative to the acquired data. Default is 50% (0.5) which results in a reference point in the middle of the acquired data.

/DEV..../SCOPES/n/TRIGRISING

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

When set (1), enables rising edge triggering. This settings is synchronized with the settings done in the /TRIGFALLING node.

/DEV..../SCOPES/n/TRIGSLOPE

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

Sets on which slope of the trigger signal the scope should trigger. This setting is synchronized with the settings done in the /TRIGFALLING and /TRIGRISING nodes.

Allowed Values:

- | | | | |
|---|--------------|---|------------------------------------|
| 0 | "none": None | 2 | "falling": Falling edge triggered. |
|---|--------------|---|------------------------------------|

1 "rising": Rising edge triggered.

3 "both": Triggers on both the rising and falling edge.

/DEV..../SCOPES/n/TRIGSTATE

Properties: Read

Type: Integer (64 bit)

Unit: None

When 1, indicates that the trigger signal satisfies the conditions of the trigger.

/DEV..../SCOPES/n/WAVE

Properties: Read, Stream

Type: ZIScopeWave

Unit: None

Contains the scope shot data.

/DEV..../SCOPES/n/SEGMENTS/COUNT

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Specifies the number of segments to be recorded in device memory. The maximum scope shot size is given by the available memory divided by the number of segments. This functionality requires the DIG option.

/DEV..../SCOPES/n/SEGMENTS/ENABLE

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enable segmented scope recording. This allows for full bandwidth recording of scope shots with a minimum dead time between individual shots. This functionality requires the DIG option.

/DEV..../SCOPES/n/STREAM/RATE

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: Hz

Streaming Rate of the scope channels. The streaming rate can be adjusted independent from the scope sampling rate. The maximum rate depends on the interface used for transfer. Note: scope streaming requires the DIG option.

Allowed Values:

4 "3.75_MHz": 3.75 MHz

10 "58.6_kHz": 58.6 kHz

5 "1.88_MHz": 1.88 MHz

11 "29.3_kHz": 29.3 kHz

6 "938_kHz": 938 kHz

12 "14.6_kHz": 14.6 kHz

7 "469_kHz": 469 kHz

13 "7.32_kHz": 7.32 kHz

8 "234_kHz": 234 kHz

14 "3.66_kHz": 3.66 kHz

9 "117_kHz": 117 kHz

15 "1.83_kHz": 1.83 kHz

/DEV..../SCOPES/n/STREAM/SAMPLE

Properties: Read, Stream

Type: ZIScopeWave

Unit: None

Streaming node containing scope sample data. Note: scope streaming requires the DIG option.

/DEV..../SCOPES/n/TRIGGATE/ENABLE

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

If enabled the trigger will be gated by the trigger gating input signal. This feature requires the DIG option.

/DEV..../SCOPES/n/TRIGGATE/INPUTSELECT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the signal source used for trigger gating if gating is enabled. This feature requires the DIG option.

Allowed Values:

- | | | | |
|---|--|---|--|
| 0 | "trigin0_high", "trigger_input0_high": Only trigger if the Trigger Input 1 is at high level. | 2 | "trigin1_high", "trigger_input1_high": Only trigger if the Trigger Input 2 is at high level. |
| 1 | "trigin0_low", "trigger_input0_low": Only trigger if the Trigger Input 1 is at low level. | 3 | "trigin1_low", "trigger_input1_low": Only trigger if the Trigger Input 2 is at low level. |

/DEV..../SCOPES/n/TRIGHYSTERESIS/ABSOLUTE

Properties: Read, Write, Setting

Type: Double

Unit: V

Defines the voltage the source signal must deviate from the trigger level before the trigger is rearmed again. Set to 0 to turn it off. The sign is defined by the Edge setting.

/DEV..../SCOPES/n/TRIGHYSTERESIS/MODE

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the mode to define the hysteresis strength. The relative mode will work best over the full input range as long as the analog input signal does not suffer from excessive noise.

Allowed Values:

- | | | | |
|---|--|---|--|
| 0 | "absolute": Selects absolute hysteresis. | 1 | "relative": Selects a hysteresis relative to the adjusted full scale signal input range. |
|---|--|---|--|

/DEV..../SCOPES/n/TRIGHYSTERESIS/RELATIVE

Properties: Read, Write, Setting

Type: Double

Unit: None

Hysteresis relative to the adjusted full scale signal input range. A hysteresis value larger than 1 (100%) is allowed.

/DEV..../SCOPES/n/CHANNELS/n/BWLIMIT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects between sample decimation and sample averaging. Averaging avoids aliasing, but may conceal signal peaks.

Allowed Values:

0 "on": On: Selects sample averaging for sample rates lower than the maximal available sampling rate.

1 "off": OFF: Selects sample decimation for sample rates lower than the maximal available sampling rate.

/DEV..../SCOPES/n/CHANNELS/n/FULLSCALE

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

Indicates the full scale value of the scope channel.

/DEV..../SCOPES/n/CHANNELS/n/INPUTSELECT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Selects the scope input signal.

Allowed Values:

0 "sigin0", "signal_input0": Signal Input 1
1 "currin0", "current_input0": Current Input 1
2 "trigin0", "trigger_input0": Trigger Input 1
3 "trigin1", "trigger_input1": Trigger Input 2
4 "auxout0", "auxiliary_output0": Aux Output 1. Requires an installed digitizer (DIG) option.
5 "auxout1", "auxiliary_output1": Aux Output 2. Requires an installed digitizer (DIG) option.
6 "auxout2", "auxiliary_output2": Aux Output 3. Requires an installed digitizer (DIG) option.

50 "demod2_r": Demod 3 R. Requires an installed digitizer (DIG) option.
51 "demod3_r": Demod 4 R. Requires an installed digitizer (DIG) option.
64 "demod0_theta": Demod 1 Θ . Requires an installed digitizer (DIG) option.
65 "demod1_theta": Demod 2 Θ . Requires an installed digitizer (DIG) option.
66 "demod2_theta": Demod 3 Θ . Requires an installed digitizer (DIG) option.
67 "demod3_theta": Demod 4 Θ . Requires an installed digitizer (DIG) option.

7	"auxout3", "auxiliary_output3": Aux Output 4. Requires an installed digitizer (DIG) option.	80	"pid0_value": PID 1 value. Requires an installed digitizer (DIG) option.
8	"auxin0", "auxiliary_input0": Aux Input 1	81	"pid1_value": PID 2 value. Requires an installed digitizer (DIG) option.
9	"auxin1", "auxiliary_input1": Aux Input 2	82	"pid2_value": PID 3 value. Requires an installed digitizer (DIG) option.
10	"demod1": Osc Φ Demod 2	83	"pid3_value": PID 4 value. Requires an installed digitizer (DIG) option.
11	"demod3": Osc Φ Demod 4	144	"pid0_shift": PID 1 Shift. Requires an installed digitizer (DIG) option.
14	"trigout0", "trigger_output0": Trigger Output 1	145	"pid1_shift": PID 2 Shift. Requires an installed digitizer (DIG) option.
15	"trigout1", "trigger_output1": Trigger Output 2	146	"pid2_shift": PID 3 Shift. Requires an installed digitizer (DIG) option.
16	"demod0_x": Demod 1 X. Requires an installed digitizer (DIG) option.	147	"pid3_shift": PID 4 Shift. Requires an installed digitizer (DIG) option.
17	"demod1_x": Demod 2 X. Requires an installed digitizer (DIG) option.	160	Reserved for future use.
18	"demod2_x": Demod 3 X. Requires an installed digitizer (DIG) option.	161	Reserved for future use.
19	"demod3_x": Demod 4 X. Requires an installed digitizer (DIG) option.	208	"pid0_error": PID 1 Error. Requires an installed digitizer (DIG) option.
32	"demod0_y": Demod 1 Y. Requires an installed digitizer (DIG) option.	209	"pid1_error": PID 2 Error. Requires an installed digitizer (DIG) option.
33	"demod1_y": Demod 2 Y. Requires an installed digitizer (DIG) option.	210	"pid2_error": PID 3 Error. Requires an installed digitizer (DIG) option.
34	"demod2_y": Demod 3 Y. Requires an installed digitizer (DIG) option.	211	"pid3_error": PID 4 Error. Requires an installed digitizer (DIG) option.
35	"demod3_y": Demod 4 Y. Requires an installed digitizer (DIG) option.	240	Reserved for future use.
48	"demod0_r": Demod 1 R. Requires an installed digitizer (DIG) option.	241	Reserved for future use.
49	"demod1_r": Demod 2 R. Requires an installed digitizer (DIG) option.	242	Reserved for future use.
		243	Reserved for future use.

/DEV..../SCOPES/n/CHANNELS/n/LIMITLOWER

Properties: Read, Write, Setting Type: Double Unit: Dependent

Lower limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.

/DEV..../SCOPES/n/CHANNELS/n/LIMITUPPER

Properties: Read, Write, Setting Type: Double Unit: Dependent

Upper limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.

/DEV..../SCOPES/n/CHANNELS/n/OFFSET

Properties: Read, Write, Setting

Type: Double

Unit: Dependent

Indicates the offset value of the scope channel.

/DEV..../SCOPES/n/STREAM/ENABLES/n

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enable scope streaming for the specified channel. This allows for continuous recording of scope data on the plotter and streaming to disk. Note: scope streaming requires the DIG option.

7.2.14. SIGINS

/DEV..../SIGINS/n/AC

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.

Allowed Values:

0 "dc": OFF: DC coupling

1 "ac": ON: AC coupling

/DEV..../SIGINS/n/AUTORANGE

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Automatic adjustment of the Range to about two times the maximum signal input amplitude measured over about 100 ms.

/DEV..../SIGINS/n/DIFF

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Switches between single ended (OFF) and differential (ON) measurements.

Allowed Values:

0 "off": OFF: Single ended voltage input

1 "on": ON: Differential voltage input

/DEV..../SIGINS/n/FLOAT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Switches the input between floating (ON) and connected to ground (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before

connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

Allowed Values:

```
0  "off": OFF: GND connected
```

```
1  "on": ON: Floating
```

```
/DEV.... /SIGINS/n/IMP50
```

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Switches between 50 Ohm (ON) and 10 M Ohm (OFF).

Allowed Values:

```
0  "10_MOhm": OFF: 10 M Ohm
```

```
1  "50_0hm": ON: 50 0hm
```

```
/DEV.... /SIGINS/n/MAX
```

Properties: Read, Write

Type: Double

Unit: V

Indicates the maximum measured value at the input.

```
/DEV.... /SIGINS/n/MIN
```

Properties: Read, Write

Type: Double

Unit: V

Indicates the minimum measured value at the input.

```
/DEV.... /SIGINS/n/ON
```

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Enables the signal input.

```
/DEV.... /SIGINS/n/RANGE
```

Properties: Read, Write, Setting

Type: Double

Unit: V

Defines the gain of the analog input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.

```
/DEV.... /SIGINS/n/SCALING
```

Properties: Read, Write, Setting

Type: Double

Unit: None

Applies the given scaling factor to the input signal.

/DEV..../SIGINS/n/RANGESTEP/TRIGGER

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Switches to the next appropriate input range such that the range fits best with the measured input signal amplitude.

7.2.15. SIGOUTS

/DEV..../SIGOUTS/n/ADD

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

The signal supplied to the Aux Input 1 is added to the signal output. For differential output the added signal is a common mode offset.

/DEV..../SIGOUTS/n/AUTORANGE

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

If enabled, selects the most suited output range automatically.

/DEV..../SIGOUTS/n/DIFF

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Switch between single-ended output (OFF) and differential output (ON). In differential mode the signal swing is defined between Signal Output +V / -V.

/DEV..../SIGOUTS/n/IMP50

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the load impedance between 50 Ohm and HiZ. The impedance of the output is always 50 Ohm. For a load impedance of 50 Ohm the displayed voltage is half the output voltage to reflect the voltage seen at the load.

Allowed Values:

0 "high_impedance": HiZ

1 "50_Ohm": 50 Ohm

/DEV..../SIGOUTS/n/OFFSET

Properties: Read, Write, Setting

Type: Double

Unit: V

Defines the DC voltage that is added to the dynamic part of the output signal.

`/DEV..../SIGOUTS/n/ON`

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enabling/Disabling the Signal Output. Corresponds to the blue LED indicator on the instrument front panel.

`/DEV..../SIGOUTS/n/OVER`

Properties: Read Type: Integer (64 bit) Unit: None

Indicates that the signal output is overloaded.

`/DEV..../SIGOUTS/n/RANGE`

Properties: Read, Write, Setting Type: Double Unit: V

Sets the output voltage range. The instrument selects the next higher available range.

`/DEV..../SIGOUTS/n/AMPLITUDES/n`

Properties: Read, Write, Setting Type: Double Unit: V

Sets the peak amplitude that the oscillator assigned to the given demodulation channel contributes to the signal output.

`/DEV..../SIGOUTS/n/ENABLES/n`

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: V

Enables individual output signal amplitude. When the MD option is used, it is possible to generate signals being the linear combination of the available demodulator frequencies.

7.2.16. STATS

`/DEV..../STATS/CPULOAD`

Properties: Read Type: Double Unit: %

Indicates the total CPU load on the machine where the data server is running.

`/DEV..../STATS/DATASERVERCPULOAD`

Properties: Read Type: Double Unit: %

Indicates the CPU load of the data server.

```
/DEV..../STATS/DATASERVERMEMLOAD
```

Properties: Read Type: Double Unit: %

Indicates the memory usage of the data server.

```
/DEV..../STATS/MEMLOAD
```

Properties: Read Type: Double Unit: %

Indicates the total memory usage of the machine where the data server is running.

/DEV..../STATS/CMDSTREAM/BANDWIDTH

Properties: Read

Type: Double

Unit: Mbit/s

Command streaming bandwidth usage on the physical network connection between device and data server.

/DEV..../STATS/CMDSTREAM/BYTESRECEIVED

Properties: Read	Type: Integer (64 bit)	Unit: B
------------------	------------------------	---------

Number of bytes received on the command stream from the device since session start.

```
/DEV..../STATS/CMDSTREAM/BYTESSENT
```

Properties: Read	Type: Integer (64 bit)	Unit: B
------------------	------------------------	---------

Number of bytes sent on the command stream from the device since session start.

/DEV..../STATS/CMDSTREAM/PACKETSLOST

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of command packets lost since device start. Command packets contain device settings that are sent to and received from the device.

/DEV..../STATS/CMDSTREAM/PACKETSRECEIVED

Properties: Read Type: Integer (64 bit) Unit: None

Number of packets received on the command stream from the device since session start.

/DEV..../STATS/CMDSTREAM/PACKETSSSENT

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of packets sent on the command stream to the device since session start.

```
/DEV..../STATS/CMDSTREAM/PENDING
```

Properties: Read Type: Integer (64 bit) Unit: None

Number of buffers ready for receiving command packets from the device.

/DEV..../STATS/CMDSTREAM/PROCESSING

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of buffers being processed for command packets. Small values indicate proper performance. For a TCP/IP interface, command packets are sent using the TCP protocol.

/DEV..../STATS/DATASTREAM/BANDWIDTH

Properties: Read Type: Double Unit: Mbit/s

Data streaming bandwidth usage on the physical network connection between device and data server.

```
/DEV..../STATS/DATASTREAM/BYTESRECEIVED
```

Properties: Read Type: Integer (64 bit) Unit: B

Number of bytes received on the data stream from the device since session start.

```
/DEV..../STATS/DATASTREAM/PACKETSLOST
```

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of data packets lost since device start. Data packets contain measurement data.

/DEV..../STATS/DATASTREAM/PACKETSRECEIVED

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of packets received on the data stream from the device since session start.

```
/DEV..../STATS/DATASTREAM/PENDING
```

Properties: Read	Type: Integer (64 bit)	Unit: None
------------------	------------------------	------------

Number of buffers ready for receiving data packets from the device.

```
/DEV..../STATS/DATASTREAM/PROCESSING
```

Properties: Read Type: Integer (64 bit) Unit: None

Number of buffers being processed for data packets. Small values indicate proper performance. For a TCP/IP interface, data packets are sent using the UDP protocol.

```
/DEV..../STATS/PHYSICAL/FANSPEED
```

Properties: Read Type: Double Unit: RPM

Speed of the internal cooling fan.

/DEV..../STATS/PHYSICAL/OVERTEMPERATURE

Properties: Read Type: Integer (64 bit) Unit: None

This flag is set to 1 if the temperature of the FPGA exceeds 85°C. It will be reset to 0 after a restart of the device.

```
/DEV..../STATS/PHYSICAL/FPGA/AUX
```

Properties: Read Type: Double Unit: V

Supply voltage of the FPGA.

```
/DEV..../STATS/PHYSICAL/FPGA/CORE
```

Properties: Read Type: Double Unit: V

Core voltage of the FPGA.

```
/DEV..../STATS/PHYSICAL/FPGA/TEMP
```

Properties: Read Type: Double Unit: °C

Internal temperature of the FPGA.

/DEV..../STATS/PHYSICAL/TEMPERATURES/n

Type: Double

Unit: °C

Internal temperature measurements.

```
/DEV..../STATS/PHYSICAL/VOLTAGES/n
```

Type: Double

Unit: V

Internal voltage measurements.

7.2.17. STATUS

```
/DEV..../STATUS/ADCOMAX
```

Type: Integer (64 bit)

Unit: None

The maximum value on Signal Input 1 (ADC0) during 100 ms.

```
/DEV..../STATUS/ADCOMIN
```

Type: Integer (64 bit)

Unit: None

The minimum value on Signal Input 1 (ADC0) during 100 ms

```
/DEV..../STATUS/ADC1MAX
```

Type: Integer (64 bit)

Unit: None

The maximum value on Signal Input 2 (ADC1) during 100 ms.

```
/DEV..../STATUS/ADC1MIN
```

Type: Integer (64 bit)

Unit: None

The minimum value on Signal Input 2 (ADC1) during 100 ms

```
/DEV..../STATUS/FIFOLEVEL
```

Type: Double

Unit: None

USB FIFO level: Indicates the USB FIFO fill level inside the device. When 100%, data is lost

```
/DEV..../STATUS/TIME
```

Properties: Read Type: Integer (64 bit) Unit: None

The current timestamp.

```
/DEV..../STATUS/FLAGS/BINARY
```

Properties: Read Type: Integer (64 bit) Unit: None

A set of binary flags giving an indication of the state of various parts of the device. Bit 0: Signal Input 1 overflow, Bit 1: Current Input 1 overflow, Bit 2: Analog PLL fail, Bit 3: Output 1 DAC OK, Bit 4: Output 2 DAC OK, Bit 5: Signal Output 1 clipping, Bit 6: Signal Output 2 clipping, Bit 7: Ext Ref 1 Locked, Bit 8: Ext Ref 2 Locked, Bit 9: Ext Ref 3 Locked, Bit 10: Ext Ref 4 Locked, Bit 11: Sample Loss, Bits 12 - 13: Trigger In 1, Bits 14 - 15: Trigger In 2, Bits 16 - 17: Trigger In 3, Bits 18 - 19: Trigger In 4, Bit 20: PLL 1 locked, Bit 21: PLL 2 locked, Bit 22: PLL 3 locked, Bit 23: PLL 4 locked, Bit 24: Rubidium clock locked, Bit 25: AU Cartesian 1 Overflow, Bit 26: AU Cartesian 2 Overflow, Bit 27: AU Polar 1 Overflow, Bit 28: AU Polar 2 Overflow.

/DEV..../STATUS/FLAGS/PACKETLOSSTCP

Properties: Read Type: Integer (64 bit) Unit: None

Flag indicating if tcp packages have been lost.

/DEV..../STATUS/FLAGS/PACKETLOSSUDP

Properties: Read Type: Integer (64 bit) Unit: None

Flag indicating if udp packages have been lost.

7.2.18. SYSTEM

```
/DEV.../SYSTEM/ACTIVEINTERFACE
```

Properties: Read Type: String Unit: None

Currently active interface of the device.

```
/DEV..../SYSTEM/EXTCLK
```

Properties: Read, Write, Setting Type: Integer (enumerated) Unit: None

10 MHz reference clock source.

Allowed Values:

0 "internal": Internal 10 MHz clock is used as the frequency and time base reference.

1 "external": An external 10 MHz clock is used as the frequency and time base reference. Provide a clean and stable 10 MHz reference to the appropriate back panel connector.

/DEV..../SYSTEM/FPGAREVISION

Properties: Read

Type: Integer (64 bit)

Unit: None

HDL firmware revision

/DEV..../SYSTEM/FWLOG

Properties: Read

Type: String

Unit: None

Returns log output of the firmware.

/DEV..../SYSTEM/FWLOGENABLE

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Enables logging to the fwlog node.

/DEV..../SYSTEM/FWREVISION

Properties: Read

Type: Integer (64 bit)

Unit: None

Revision of the device internal controller software

/DEV..../SYSTEM/IDENTIFY

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Setting this node to 1 will cause the device to blink the power led for a few seconds.

/DEV..../SYSTEM/INTERFACESPEED

Properties: Read

Type: String

Unit: None

Speed of the currently active interface (USB only).

/DEV..../SYSTEM/JUMBO

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Enables jumbo frames (4k) on the TCP/IP interface. This will reduce the load on the PC and is required to achieve maximal throughput. Make sure that jumbo frames (4k) are enabled on the

network card as well. If one of the devices on the network is not able to work with jumbo frames, the connection will fail.

```
/DEV.... /SYSTEM/OWNER
```

Properties: Read Type: String Unit: None

Returns the current owner of the device (IP).

```
/DEV.... /SYSTEM/PORTTCP
```

Properties: Read, Write Type: Integer (64 bit) Unit: None

Returns the current TCP port used for communication to the dataserver.

```
/DEV.... /SYSTEM/PORTUDP
```

Properties: Read, Write Type: Integer (64 bit) Unit: None

Returns the current UDP port used for communication to the dataserver.

```
/DEV..../SYSTEM/PREAMPENABLE
```

Properties: Read, Write, Setting Type: Integer (64 bit) Unit: None

Enables the preamplifier.

```
/DEV.... /SYSTEM/SAVEPORTS
```

Properties: Read, Write Type: Integer (64 bit) Unit: None

Flag indicating that the TCP and UDP ports should be saved.

```
/DEV.... /SYSTEM/SHUTDOWN
```

Properties: Read, Write Type: Integer (64 bit) Unit: None

Sending a '1' to this node initiates a shutdown of the operating system on the MFLI device. It is recommended to trigger this shutdown before switching the device off with the hardware switch at the back side of the device.

```
/DEV..../SYSTEM/STALL
```

Properties: Read, Write	Type: Integer (64 bit)	Unit: None
-------------------------	------------------------	------------

Indicates if the network connection is stalled.

/DEV..../SYSTEM/BOARDREVISIONS/n

Properties: Read

Type: String

Unit: None

Hardware revision of the FPGA base board

/DEV..../SYSTEM/COMPDELAY/CALIBRATE

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Perform automatic calibration of the input delay compensation.

/DEV..../SYSTEM/IMPEDANCE/APPLICATION

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the Impedance application

Allowed Values:

0 "lcr": LCR Impedance Measurement:
General-purpose settings for measuring
components at finite frequency.

1 "dc": DC Impedance Measurement:
Settings for measuring resistance at zero
frequency.

/DEV..../SYSTEM/IMPEDANCE/FILTER

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Switch between application-based or manually configured impedance settings. A parameter set given by an Application mode can be fine-tuned by changing to Advanced mode. Changing back to Application mode will reset the parameters.

Allowed Values:

0 "application": Application: The
impedance settings are adjusted to fit
best the selected application.

1 "advanced": Advanced: The impedance
settings are manually configured.

/DEV..../SYSTEM/IMPEDANCE/PRECISION

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the Impedance precision and measurement speed by automatically adjusting the filter bandwidth. If the sweeper module is in impedance application mode the precision setting will also control the sweeper measurement speed.

Allowed Values:

0 "low": Low -> fast settling: Medium
accuracy/precision is optimized for fast
response.

2 "very_high": Very high -> slow settling:
Very high accuracy/precision uses long
settling time due to very low bandwidth.

- ```
/DEV..../SYSTEM/PRESET/BUSY
```

Unit: None

```
/DEV..../SYSTEM/PRESET/DEFAULT
```

Unit: None

- 3 Select the preset stored in internal flash memory at position 3 as default preset.

## Unit: None

Unit: None

Unit: None

- 4 Select the preset stored in internal flash memory at position 4.

- |   |                                                                  |   |                                                                  |
|---|------------------------------------------------------------------|---|------------------------------------------------------------------|
| 1 | Select the preset stored in internal flash memory at position 1. | 5 | Select the preset stored in internal flash memory at position 5. |
| 2 | Select the preset stored in internal flash memory at position 2. | 6 | Select the preset stored in internal flash memory at position 6. |
| 3 | Select the preset stored in internal flash memory at position 3. |   |                                                                  |

### /DEV..../SYSTEM/PRESET/LOAD

|                         |                        |            |
|-------------------------|------------------------|------------|
| Properties: Read, Write | Type: Integer (64 bit) | Unit: None |
|-------------------------|------------------------|------------|

Load the selected preset.

### /DEV..../SYSTEM/PRESET/SAVE

|                         |                        |            |
|-------------------------|------------------------|------------|
| Properties: Read, Write | Type: Integer (64 bit) | Unit: None |
|-------------------------|------------------------|------------|

Save the actual setting as preset.

### /DEV..../SYSTEM/PROPERTIES/FREQRESOLUTION

|                  |                        |            |
|------------------|------------------------|------------|
| Properties: Read | Type: Integer (64 bit) | Unit: None |
|------------------|------------------------|------------|

The number of bits used to represent a frequency.

### /DEV..../SYSTEM/PROPERTIES/FREQSCALING

|                  |              |            |
|------------------|--------------|------------|
| Properties: Read | Type: Double | Unit: None |
|------------------|--------------|------------|

The scale factor to use to convert a frequency represented as a freqresolution-bit integer to a floating point value.

### /DEV..../SYSTEM/PROPERTIES/MAXFREQ

|                  |              |            |
|------------------|--------------|------------|
| Properties: Read | Type: Double | Unit: None |
|------------------|--------------|------------|

The maximum oscillator frequency that can be set.

### /DEV..../SYSTEM/PROPERTIES/MAXTIMECONSTANT

|                  |              |            |
|------------------|--------------|------------|
| Properties: Read | Type: Double | Unit: None |
|------------------|--------------|------------|

The maximum demodulator time constant that can be set.

### /DEV..../SYSTEM/PROPERTIES/MINFREQ

|                  |              |            |
|------------------|--------------|------------|
| Properties: Read | Type: Double | Unit: None |
|------------------|--------------|------------|

The minimum oscillator frequency that can be set.

```
/DEV..../SYSTEM/PROPERTIES/MINTIMECONSTANT
```

Unit: None

The minimum demodulator time constant that can be set.

## /DEV..../SYSTEM/PROPERTIES/NEGATIVEFREQ

Unit: None

Indicates whether negative frequencies are supported.

```
/DEV..../SYSTEM/PROPERTIES/TIMEBASE
```

Unit: s

Minimal time difference between two timestamps. Is equal to  $1/(\text{maximum sampling rate})$ .

```
/DEV..../SYSTEM/COMPDELAY/DELAYS/n
```

Unit: None

Current values of the compensation delays. 0: Signal Input 0, 1: Signal Input 1, 2: Aux Inputs, 3: Trigger Inputs, 4: Loopbacks

## /DEV..../SYSTEM/IMPEDANCE/CALIB/CABLELENGTH

Unit: None

Length of cables (One way) used in measurement set up.

Allowed Values:

3 "3\_m": Cable length is set to 3 m.

1 "1\_m": Cable length is set to 1 m.

```
/DEV..../SYSTEM/NICS/n/DEFAULTGATEWAY
```

Unit: None

Default gateway configuration for the network connection.

```
/DEV..../SYSTEM/NICS/n/DEFAULTIP4
```

Unit: None

IPv4 address of the device to use if static IP is enabled.

#### /DEV..../SYSTEM/NICS/n/DEFAULTMASK

Properties: Read, Write                      Type: String                      Unit: None

IPv4 mask in case of static IP.

#### /DEV..../SYSTEM/NICS/n/GATEWAY

Properties: Read                              Type: String                      Unit: None

Current network gateway.

#### /DEV..../SYSTEM/NICS/n/IP4

Properties: Read                              Type: String                      Unit: None

Current IPv4 of the device.

#### /DEV..../SYSTEM/NICS/n/JUMBO

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Stored setting for jumbo frames. Will be applied after reboot.

#### /DEV..../SYSTEM/NICS/n/MAC

Properties: Read                              Type: String                      Unit: None

Current MAC address of the device network interface.

#### /DEV..../SYSTEM/NICS/n/MASK

Properties: Read                              Type: String                      Unit: None

Current network mask.

#### /DEV..../SYSTEM/NICS/n/SAVEIP

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

If written, this action will program the defined static IP address to the device.

#### /DEV..../SYSTEM/NICS/n/STATIC

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Enable this flag if the device is used in a network with fixed IP assignment without a DHCP server.

```
/DEV..../SYSTEM/UPDATE/LABONE/BUSY
```

Properties: Read      Type: Integer (64 bit)      Unit: None

Device is updating.

## /DEV..../SYSTEM/UPDATE/LABONE/DATA

Properties: Read, Write      Type: ZIVectorData      Unit: None

Node which accepts the Binary file to update the device.

```
/DEV.... /SYSTEM/UPDATE/LABONE/DONE
```

Properties: Read      Type: Integer (64 bit)      Unit: None

Updating has finished.

## /DEV..../SYSTEM/UPDATE/LABONE/ERROR

Properties: Read      Type: Integer (64 bit)      Unit: None

An error occurred during the update.

```
/DEV.... /SYSTEM/UPDATE/LABONE/PROGRESS
```

Properties: Read                      Type: Double                      Unit: None

A value between 0 and 1 indicating the update progress.

```
/DEV.... /SYSTEM/UPDATE/LABONE/START
```

Properties: Read, Write      Type: Integer (64 bit)      Unit: None

If set to 1 the device starts the update.

```
/DEV.../SYSTEM/IMPEDANCE/CALIB/INTERNAL/
ALLOWSMOOTHING
```

Properties: Read, Write      Type: Integer (64 bit)      Unit: None

Indicates whether smoothing can be applied to the device internal compensation data.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/INTERNAL/DATA

Properties: Read, Write                      Type: ZIVectorData                      Unit: None

Contains the user compensation data in binary form.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/INTERNAL/INFO

Properties: Read                              Type: String                              Unit: None

Contains general information about the device internal compensation data.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/INTERNAL/STORE

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Store the compensation data persistently on the device. The compensation will be loaded automatically on every power-up.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/INTERNAL/VALID

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Indicates whether the device internal calibration data is valid (1 = valid, 0 = invalid).

### /DEV..../SYSTEM/IMPEDANCE/CALIB/USER/ALLOWSMOOTHING

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Indicates whether smoothing can be applied to the user calibration data.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/USER/DATA

Properties: Read, Write                      Type: ZIVectorData                      Unit: None

Contains the user compensation data in binary form.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/USER/INFO

Properties: Read                              Type: String                              Unit: None

Contains general information about the user compensation data.

### /DEV..../SYSTEM/IMPEDANCE/CALIB/USER/VALID

Properties: Read, Write                      Type: Integer (64 bit)                      Unit: None

Indicates whether the user compensation data is valid (1 = valid, 0 = invalid).

/DEV..../SYSTEM/IMPEDANCE/FREQLIMIT2T/n/FREQ

Properties: Read                      Type: Double                      Unit: None

Indicates the limit of a frequency range for 2-terminal impedance measurements.

/DEV..../SYSTEM/IMPEDANCE/FREQLIMIT2T/n/RANGE

Properties: Read      Type: Double      Unit: None

Indicates the minimum input current range that can be used at frequencies at or above the value in the corresponding freq node for 2-terminal impedance measurements.

/DEV..../SYSTEM/IMPEDANCE/FREQLIMIT4T/n/FREQ

Properties: Read                      Type: Double                      Unit: None

Indicates the limit of a frequency range for 4-terminal impedance measurements.

/DEV..../SYSTEM/IMPEDANCE/FREQLIMIT4T/n/RANGE

Properties: Read                      Type: Double                      Unit: None

Indicates the minimum input current range that can be used at frequencies at or above the value in the corresponding freq node for 4-terminal impedance measurements.

/DEV..../SYSTEM/PRESET/RECORDS/n/FEATURES

Properties: Read      Type: Integer (64 bit)      Unit: None

### Properties of the preset.

/DEV..../SYSTEM/PRESET/RECORDS/n/LABEL

Properties: Read, Write      Type: String      Unit: None

Name of this preset.

/DEV..../SYSTEM/PRESET/RECORDS/n/TIMESTAMP

|                         |                        |            |
|-------------------------|------------------------|------------|
| Properties: Read, Write | Type: Integer (64 bit) | Unit: None |
|-------------------------|------------------------|------------|

Not used.

### /DEV..../SYSTEM/PRESET/RECORDS/n/VALID

Properties: Read

Type: Integer (64 bit)

Unit: None

True if a valid preset is stored.

## 7.2.19. TRIGGERS

### /DEV..../TRIGGERS/IN/n/AUTOTHRESHOLD

Properties: Read, Write

Type: Integer (64 bit)

Unit: None

Automatically adjust the trigger threshold. The level is adjusted to fall in the center of the applied transitions.

### /DEV..../TRIGGERS/IN/n/LEVEL

Properties: Read, Write, Setting

Type: Double

Unit: V

Trigger voltage level at which the trigger input toggles between low and high. Use 50% amplitude for digital input and consider the trigger hysteresis.

### /DEV..../TRIGGERS/OUT/n/PULSEWIDTH

Properties: Read, Write, Setting

Type: Double

Unit: s

Defines the minimal pulse width for the case of Scope events written to the trigger outputs of the device.

### /DEV..../TRIGGERS/OUT/n/SOURCE

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the signal assigned to the trigger output.

Allowed Values:

|   |                                                                                                               |    |                                                             |
|---|---------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------|
| 0 | "disabled": The output trigger is disabled.                                                                   | 7  | "scope_not_active": Scope /Active. Requires the DIG Option. |
| 1 | "demod1_phase": Oscillator phase of demod 2 (trigger output channel 1) or demod 4 (trigger output channel 2). | 36 | "th_logic_unit0_out": Threshold Logic Unit 1 output.        |
| 2 | "scope_trigger": Scope Trigger. Requires the DIG Option.                                                      | 37 | "th_logic_unit1_out": Threshold Logic Unit 2 output.        |
| 3 | "scope_not_trigger": Scope /Trigger. Requires the DIG Option.                                                 | 38 | "th_logic_unit2_out": Threshold Logic Unit 3 output.        |
| 4 | "scope_armed": Scope Armed. Requires the DIG Option.                                                          | 39 | "th_logic_unit3_out": Threshold Logic Unit 4 output.        |



- |   |                                                           |    |                               |
|---|-----------------------------------------------------------|----|-------------------------------|
| 5 | "scope_not_armed": Scope /Armed. Requires the DIG Option. | 52 | "mds_sync_out": MDS Sync Out. |
| 6 | "scope_active": Scope Active. Requires the DIG Option.    |    |                               |

## 7.2.20. TU (Threshold Unit)

/DEV..../TU/LOGICUNITS/n/INTERNALVALUE

|                  |                        |            |
|------------------|------------------------|------------|
| Properties: Read | Type: Integer (64 bit) | Unit: None |
|------------------|------------------------|------------|

Show the value after logical combination but before output settings. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

/DEV..../TU/LOGICUNITS/n/INVERT

|                                  |                        |            |
|----------------------------------|------------------------|------------|
| Properties: Read, Write, Setting | Type: Integer (64 bit) | Unit: None |
|----------------------------------|------------------------|------------|

Changes the output signal to low-active.

/DEV..../TU/LOGICUNITS/n/KEEP

|                                  |                        |            |
|----------------------------------|------------------------|------------|
| Properties: Read, Write, Setting | Type: Integer (64 bit) | Unit: None |
|----------------------------------|------------------------|------------|

Hold the output state indefinitely once it has changed to the activated state.

/DEV..../TU/LOGICUNITS/n/PULSEWIDTH

|                                  |              |         |
|----------------------------------|--------------|---------|
| Properties: Read, Write, Setting | Type: Double | Unit: s |
|----------------------------------|--------------|---------|

Minimum pulse width for the generated output signal. The width of pulses shorter than this time will be extended.

/DEV..../TU/LOGICUNITS/n/VALUE

|                  |                        |            |
|------------------|------------------------|------------|
| Properties: Read | Type: Integer (64 bit) | Unit: None |
|------------------|------------------------|------------|

Shows the output value of the logic unit. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

/DEV..../TU/THRESHOLDS/n/ACTIVATIONTIME

|                                  |              |         |
|----------------------------------|--------------|---------|
| Properties: Read, Write, Setting | Type: Double | Unit: s |
|----------------------------------|--------------|---------|

Minimum duration of threshold violation needed before the output is activated. This can be used as a glitch filter.

## /DEV..../TU/THRESHOLDS/n/DEACTIVATIONTIME

Properties: Read, Write, Setting

Type: Double

Unit: s

Minimum duration of threshold compliance needed before the output is deactivated.

## /DEV..../TU/THRESHOLDS/n/INPUT

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Select the signal source to be used in the threshold unit.

Allowed Values:

|    |                              |    |                                                                           |
|----|------------------------------|----|---------------------------------------------------------------------------|
| 0  | "demod_x": Demod X           | 50 | "dio": DIO                                                                |
| 1  | "demod_y": Demod Y           | 51 | "trigin", "trigger_input": Trigger In                                     |
| 2  | "demod_r": Demod R           | 52 | "trigout", "trigger_output": Trigger Out                                  |
| 3  | "demod_theta": Demod Theta   | 53 | "input_overload_v": Input Overload (V)                                    |
| 5  | "pid_out": PID Out           | 54 | "input_overload_i": Input Overload (I)                                    |
| 6  | "boxcar": Boxcar             | 55 | "output_overload": Output Overload                                        |
| 7  | "au_cartesian": AU Cartesian | 56 | "auxin_overload",<br>"auxiliary_input_overload": Aux input<br>Overload    |
| 8  | "au_polar": AU Polar         | 57 | "auxout_overload",<br>"auxiliary_output_overload": Aux<br>Output Overload |
| 9  | "pid_shift": PID Shift       | 58 | "pid_output_overload": PID Output<br>Overload                             |
| 10 | "pid_error": PID Error       | 59 | "tu_output": TU Output Value                                              |

## /DEV..../TU/THRESHOLDS/n/INPUTCHANNEL

Properties: Read, Write, Setting

Type: Integer (64 bit)

Unit: None

Select the channel according to the selected signal source.

## /DEV..../TU/THRESHOLDS/n/INTERNALVALUE

Properties: Read

Type: Integer (64 bit)

Unit: Dependent

Shows the value after threshold but before minimum time. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

## /DEV..../TU/THRESHOLDS/n/VALUE

Properties: Read

Type: Integer (64 bit)

Unit: Dependent

Shows the threshold value at the threshold output. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

### /DEV..../TU/LOGICUNITS/n/INPUTS/n/CHANNEL

Properties: Read, Write, Setting      Type: Integer (64 bit)      Unit: None

Selects threshold channel to be used as input.

### /DEV..../TU/LOGICUNITS/n/INPUTS/n/NOT

Properties: Read, Write, Setting      Type: Integer (64 bit)      Unit: None

Inverts the input signal.

### /DEV..../TU/LOGICUNITS/n/OPS/n/VALUE

Properties: Read, Write, Setting      Type: Integer (enumerated)      Unit: None

Logical operation applied to the signals to the left and right of the control.

Allowed Values:

- |                                                                                                            |                                         |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 0   "off": No logical operation selected.<br>No additional signals will be used to<br>generate the output. | 2   "or": Use a logical OR operation.   |
| 1   "and": Use a logical AND operation.                                                                    | 3   "xor": Use a logical XOR operation. |

### /DEV..../TU/THRESHOLDS/n/FILTER/ABS

Properties: Read, Write, Setting      Type: Integer (64 bit)      Unit: None

Takes the absolute value of an analog input signal.

### /DEV..../TU/THRESHOLDS/n/FILTER/ENABLE

Properties: Read      Type: Integer (64 bit)      Unit: None

Enables the low-pass filter applied to the analog input signal

### /DEV..../TU/THRESHOLDS/n/FILTER/FILTEREDVALUE

Properties: Read      Type: Double      Unit: Dependent

Shows the value after the low-pass filter.

### /DEV..../TU/THRESHOLDS/n/FILTER/MODE

Properties: Read, Write, Setting      Type: Integer (enumerated)      Unit: None

Selects the analysis mode defining the output signal.

Allowed Values:

- |   |                                                                                                |   |                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | "above_upper": Enable if the Signal is above Upper Threshold.                                  | 3 | "crossing_both_rising": Enable if the Signal crosses Upper Threshold from below Lower Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis.  |
| 1 | "below_lower": Enable if the Signal is below Lower Threshold.                                  | 4 | "crossing_both_falling": Enable if the Signal crosses Lower Threshold from above Upper Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis. |
| 2 | "outside_range": Enable if the Signal is outside the range [Lower Threshold, Upper Threshold]. |   |                                                                                                                                                                                      |

### /DEV..../TU/THRESHOLDS/n/FILTER/THRESHOLDHIGH

Properties: Read, Write, Setting      Type: Double      Unit: Dependent

Upper threshold used to generate the output. In Falling Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Upper Threshold from below.

### /DEV..../TU/THRESHOLDS/n/FILTER/THRESHOLDLOW

Properties: Read, Write, Setting      Type: Double      Unit: Dependent

Lower threshold used to generate the output. In Rising Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Lower Threshold from above.

### /DEV..../TU/THRESHOLDS/n/FILTER/TIMECONSTANT

Properties: Read, Write, Setting      Type: Double      Unit: s

Defines the characteristic time constant (cut off) of the low-pass filter applied to the analog input signal.

## 7.2.21. ZI (Labone Data Server Nodes)

### /ZI/ABOUT/COMMIT

Properties: Read      Type: String      Unit: None

Contains the commit hash of the source code used to build this version of the LabOne software.

### /ZI/ABOUT/COPYRIGHT

Properties: Read      Type: String      Unit: None

Holds the copyright notice.

/ZI/ABOUT/DATASERVER

Properties: Read

Type: String

Unit: None

Contains information about the Zurich Instruments Data Server.

## /ZI/ABOUT/FWREVISION

Properties: Read

Type: Integer (64 bit)

Unit: None

Contains the revision of the device firmware.

## /71/ABOUT/REVISION

Properties: Read

Type: Integer (64 bit)

Unit: None

Contains the revision number of the Zurich Instruments Data Server.

## /ZI/ABOUT/VERSION

Properties: Read

Type: String

Unit: None

Contains the version of the LabOne software.

## /ZI/CLOCKBASE

Properties: Read

Type: Double

Unit: None

A fallback clock frequency that can be used by clients for calculating time bases when no other is available.

## /ZI/CONFIG/OPEN

Properties: Read, Write, Setting

Type: Integer (enumerated)

Unit: None

Enable communication with the LabOne Data Server from other computers in the network.

Allowed Values:

- 0 "local": Communication only possible with the local machine.

1 "network": Communication possible with  
other machines in the network.

## /ZI/CONFIG/PORT

Properties: Read

Type: Integer (64 bit)

Unit: None

The IP port on which the LabOne Data Server listens.

## /ZI/DEBUG/LEVEL

Properties: Read, Write, Setting      Type: Integer (enumerated)      Unit: None

Set the logging level (amount of detail) of the LabOne Data Server.

Allowed Values:

- |   |                                                                   |   |                                                                 |
|---|-------------------------------------------------------------------|---|-----------------------------------------------------------------|
| 0 | "trace": Trace. Messages designated as traces are logged.         | 4 | "warning": Warning. Messages designated as warnings are logged. |
| 1 | "debug": Debug. Messages designated as debugging info are logged. | 5 | "error": Error. Messages designated as errors are logged.       |
| 2 | "info": Info. Messages designated as informational are logged.    | 6 | "fatal": Fatal. Messages designated as fatal errors are logged. |
| 3 | "status": Status. Messages designated as status info are logged.  |   |                                                                 |

## /ZI/DEBUG/LOG

Properties: Read      Type: String      Unit: None

Returns the logfile text of the LabOne Data Server.

## /ZI/DEBUG/LOGPATH

Properties: Read      Type: String      Unit: None

Returns the path of the log directory.

## /ZI/DEVICES/CONNECTED

Properties: Read      Type: String      Unit: None

Contains a list of devices connected to the LabOne Data Server.

## /ZI/DEVICES/DISCOVER

Properties: Read, Write      Type: String      Unit: None

Not used.

## /ZI/DEVICES/VISIBLE

Properties: Read      Type: String      Unit: None

Contains a list of devices in the network visible to the LabOne Data Server.

### /ZI/MDS/GROUPS/n/DEVICES

Properties: Read, Write, Setting      Type: String      Unit: None

Contains a list of devices in this synchronization group.

### /ZI/MDS/GROUPS/n/KEEPALIVE

Properties: Read, Write, Setting      Type: Integer (64 bit)      Unit: None

Set by the MDS module to indicate control over this synchronization group.

### /ZI/MDS/GROUPS/n/LOCKED

Properties: Read, Write, Setting      Type: Integer (64 bit)      Unit: None

Indicates whether the device group is locked by a MDS module.

### /ZI/MDS/GROUPS/n/STATUS

Properties: Read, Write, Setting      Type: Integer (enumerated)      Unit: None

Indicates the status the synchronization group.

Allowed Values:

|    |                                                          |   |       |
|----|----------------------------------------------------------|---|-------|
| -1 | Error. An error occurred in the synchronization process. | 1 | Sync  |
| 0  | New                                                      | 2 | Alive |

### /ZI/SYSTEM/USAGEDATA

Properties: Read      Type: String      Unit: None

Contains a JSON formatted string giving usage information about the LabOne Software (which tabs/modules used, error conditions).

---

# Glossary

This glossary provides easy to understand descriptions for many terms related to measurement instrumentation including the abbreviations used inside this user manual.

## A

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A/D                              | Analog to Digital<br>See Also <a href="#">ADC</a> .                                                                                                                                                                                                                                                                                                                                                                                                             |
| AC                               | Alternate Current                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ADC                              | Analog to Digital Converter                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| AM                               | Amplitude Modulation                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Amplitude Modulated AFM (AM-AFM) | AFM mode where the amplitude change between drive and measured signal encodes the topography or the measured AFM variable.<br>See Also <a href="#">Atomic Force Microscope</a> .                                                                                                                                                                                                                                                                                |
| API                              | Application Programming Interface                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ASCII                            | American Standard Code for Information Interchange                                                                                                                                                                                                                                                                                                                                                                                                              |
| Atomic Force Microscope (AFM)    | Microscope that scans surfaces by means an oscillating mechanical structure (e.g. cantilever, tuning fork) whose oscillating tip gets so close to the surface to enter in interaction because of electrostatic, chemical, magnetic or other forces. With an AFM it is possible to produce images with atomic resolution.<br>See Also <a href="#">Amplitude Modulated AFM</a> , <a href="#">Frequency Modulated AFM</a> , <a href="#">Phase modulation AFM</a> . |
| AVAR                             | Allen Variance                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## B

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bandwidth (BW) | <p>The signal bandwidth represents the highest frequency components of interest in a signal. For filters the signal bandwidth is the cut-off point, where the transfer function of a system shows 3 dB attenuation versus DC. In this context the bandwidth is a synonym of cut-off frequency <math>f_{\text{cut-off}}</math> or 3dB frequency <math>f_{-3\text{dB}}</math>. The concept of bandwidth is used when the dynamic behavior of a signal is important or separation of different signals is required.</p> <p>In the context of a open-loop or closed-loop system, the bandwidth can be used to indicate the fastest speed of the system, or the highest signal update change rate that is possible with the system.</p> <p>Sometimes the term bandwidth is erroneously used as synonym of frequency range.<br/>See Also <a href="#">Noise Equivalent Power Bandwidth</a>.</p> |
| BNC            | Bayonet Neill-Concelman Connector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## C

|    |                                               |
|----|-----------------------------------------------|
| CF | Clock Fail (internal processor clock missing) |
|----|-----------------------------------------------|



---

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Common Mode Rejection Ratio (CMRR) | Specification of a differential amplifier (or other device) indicating the ability of an amplifier to obtain the difference between two inputs while rejecting the components that do not differ from the signal (common mode). A high CMRR is important in applications where the signal of interest is represented by a small voltage fluctuation superimposed on a (possibly large) voltage offset, or when relevant information is contained in the voltage difference between two signals. The simplest mathematical definition of common-mode rejection ratio is: $CMRR = 20 * \log(\text{differential gain} / \text{common mode gain})$ . |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|     |                        |
|-----|------------------------|
| CSV | Comma Separated Values |
|-----|------------------------|

## D

|                      |                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D/A                  | Digital to Analog                                                                                                                                                                                             |
| DAC                  | Digital to Analog Converter                                                                                                                                                                                   |
| DC                   | Direct Current                                                                                                                                                                                                |
| DDS                  | Direct Digital Synthesis                                                                                                                                                                                      |
| DHCP                 | Dynamic Host Configuration Protocol                                                                                                                                                                           |
| DIO                  | Digital Input/Output                                                                                                                                                                                          |
| DNS                  | Domain Name Server                                                                                                                                                                                            |
| DSP                  | Digital Signal Processor                                                                                                                                                                                      |
| DUT                  | Device Under Test                                                                                                                                                                                             |
| Dynamic Reserve (DR) | The measure of a lock-in amplifier's capability to withstand the disturbing signals and noise at non-reference frequencies, while maintaining the specified measurement accuracy within the signal bandwidth. |

## E

|     |                                                               |
|-----|---------------------------------------------------------------|
| XML | Extensible Markup Language.<br>See Also <a href="#">XML</a> . |
|-----|---------------------------------------------------------------|

## F

|                                   |                                                                                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FFT                               | Fast Fourier Transform                                                                                                                                                           |
| FIFO                              | First In First Out                                                                                                                                                               |
| FM                                | Frequency Modulation                                                                                                                                                             |
| Frequency Accuracy (FA)           | Measure of an instrument's ability to faithfully indicate the correct frequency versus a traceable standard.                                                                     |
| Frequency Modulated AFM (FM-AFM)  | AFM mode where the frequency change between drive and measured signal encodes the topography or the measured AFM variable.<br>See Also <a href="#">Atomic Force Microscope</a> . |
| Frequency Response Analyzer (FRA) | Instrument capable to stimulate a device under test and plot the frequency response over a selectable frequency range with a fine granularity.                                   |

---

Frequency Sweeper

See Also [Frequency Response Analyzer](#).

## G

Gain Phase Meter

See Also [Vector Network Analyzer](#).

GPIO

General Purpose Interface Bus

GUI

Graphical User Interface

## I

I/O

Input / Output

Impedance Spectroscope  
(IS)

Instrument suited to stimulate a device under test and to measure the impedance (by means of a current measurement) at a selectable frequency and its amplitude and phase change over time. The output is both amplitude and phase information referred to the stimulus signal.

Input Amplitude Accuracy  
(IAA)

Measure of instrument's capability to faithfully indicate the signal amplitude at the input channel versus a traceable standard.

Input voltage noise (IVN)

Total noise generated by the instrument and referred to the signal input, thus expressed as additional source of noise for the measured signal.

IP

Internet Protocol

## L

LAN

Local Area Network

LED

Light Emitting Diode

Lock-in Amplifier (LI, LIA)

Instrument suited for the acquisition of small signals in noisy environments, or quickly changing signal with good signal to noise ratio - lock-in amplifiers recover the signal of interest knowing the frequency of the signal by demodulation with the suited reference frequency - the result of the demodulation are amplitude and phase of the signal compared to the reference: these are value pairs in the complex plane  $(X, Y)$ ,  $(R, \Theta)$ .

## M

Media Access Control  
address (MAC address)

Refers to the unique identifier assigned to network adapters for physical network communication.

Multi-frequency (MF)

Refers to the simultaneous measurement of signals modulated at arbitrary frequencies. The objective of multi-frequency is to increase the information that can be derived from a measurement which is particularly important for one-time, non-repeating events, and to increase the speed of a measurement since different frequencies do not have to be applied one after the other.

See Also [Multi-harmonic](#).

Multi-harmonic (MH)

Refers to the simultaneous measurement of modulated signals at various harmonic frequencies. The objective of multi-frequency is to increase the

---

information that can be derived from a measurement which is particularly important for one-time, non-repeating events, and to increase the speed of a measurement since different frequencies do not have to be applied one after the other.  
See Also [Multi-frequency](#).

## N

|                                          |                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noise Equivalent Power Bandwidth (NEPBW) | Effective bandwidth considering the area below the transfer function of a low-pass filter in the frequency spectrum. NEPBW is used when the amount of power within a certain bandwidth is important, such as noise measurements. This unit corresponds to a perfect filter with infinite steepness at the equivalent frequency.<br>See Also <a href="#">Bandwidth</a> . |
| Nyquist Frequency (NF)                   | For sampled analog signals, the Nyquist frequency corresponds to two times the highest frequency component that is being correctly represented after the signal conversion.                                                                                                                                                                                             |

## O

|                                 |                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Output Amplitude Accuracy (OAA) | Measure of an instrument's ability to faithfully output a set voltage at a given frequency versus a traceable standard. |
| OV                              | Over Volt (signal input saturation and clipping of signal)                                                              |

## P

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PC                            | Personal Computer                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| PD                            | Phase Detector                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Phase-locked Loop (PLL)       | Electronic circuit that serves to track and control a defined frequency. For this purpose a copy of the external signal is generated such that it is in phase with the original signal, but with usually better spectral characteristics. It can act as frequency stabilization, frequency multiplication, or as frequency recovery. In both analog and digital implementations it consists of a phase detector, a loop filter, a controller, and an oscillator. |
| Phase modulation AFM (PM-AFM) | AFM mode where the phase between drive and measured signal encodes the topography or the measured AFM variable.<br>See Also <a href="#">Atomic Force Microscope</a> .                                                                                                                                                                                                                                                                                            |
| PID                           | Proportional-Integral-Derivative                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PL                            | Packet Loss (loss of packets of data between the instruments and the host computer)                                                                                                                                                                                                                                                                                                                                                                              |

## R

|                        |                                                                                                                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RISC                   | Reduced Instruction Set Computer                                                                                                                                                                                               |
| Root Mean Square (RMS) | Statistical measure of the magnitude of a varying quantity. It is especially useful when variates are positive and negative, e.g., sinusoids, sawtooth, square waves. For a sine wave the following relation holds between the |

---

amplitude and the RMS value:  $U_{\text{RMS}} = U_{\text{PK}} / \sqrt{2} = U_{\text{PK}} / 1.41$ . The RMS is also called quadratic mean.

RT Real-time

## S

Scalar Network Analyzer (SNA) Instrument that measures the voltage of an analog input signal providing just the amplitude (gain) information.  
See Also [Spectrum Analyzer](#), [Vector Network Analyzer](#).

SL Sample Loss (loss of samples between the instrument and the host computer)

Spectrum Analyzer (SA) Instrument that measures the voltage of an analog input signal providing just the amplitude (gain) information over a defined spectrum.  
See Also [Scalar Network Analyzer](#).

SSH Secure Shell

## T

TC Time Constant

TCP/IP Transmission Control Protocol / Internet Protocol

Thread An independent sequence of instructions to be executed by a processor.

Total Harmonic Distortion (THD) Measure of the non-linearity of signal channels (input and output)

TTL Transistor to Transistor Logic level

## U

UHF Ultra-High Frequency

UHS Ultra-High Stability

USB Universal Serial Bus

## V

VCO Voltage Controlled Oscillator

Vector Network Analyzer (VNA) Instrument that measures the network parameters of electrical networks, commonly expressed as s-parameters. For this purpose it measures the voltage of an input signal providing both amplitude (gain) and phase information. For this characteristic an older name was gain phase meter.  
See Also [Gain Phase Meter](#), [Scalar Network Analyzer](#).

## X

XML Extensible Markup Language: Markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.

---

## Z

|         |                                                                                                                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZCtrl   | Zurich Instruments Control bus                                                                                                                                                                                                                                                               |
| ZoomFFT | This technique performs FFT processing on demodulated samples, for instance after a lock-in amplifier. Since the resolution of an FFT depends on the number of point acquired and the spanned time (not the sample rate), it is possible to obtain very highly resolution spectral analysis. |
| ZSync   | Zurich Instruments Synchronization bus                                                                                                                                                                                                                                                       |

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